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We need answer user. Need act as licensed PE experienced instructor. Provide comprehensive exam-focused review only Stormwater Collection and Drainage within Hydraulics and Hydrology area of NCEES PE Civil: Construction. Must align with specs and handbook. They provided huge handbook text. We need produce review. Need likely lengthy. We need include definitions formulas step-by-step worked examples high-yield tips practice problems detailed solutions tables bullet lists concise summaries rapid review. End with high-yield checklist and self-test. Flag details not specified. Do not guarantee pass. We need cover open-channel flow, culverts, storm sewers, watersheds, construction applications, QC, common errors. We can reference PE Civil Reference Handbook v2.0 sections. Need use USCS mostly maybe SI. Need include equations. Need be careful with NCEES scope. They want only stormwater collection and drainage. We can include related closed conduit etc as apply.

We need answer likely long. Desired oververbosity 5 but task demands comprehensive. We'll produce structured review. Need no analysis. Ensure formulas correct.

Need maybe note that handbook doesn't include all info; some constants are from FHWA HDS-5/HEC-22 not necessarily supplied? In exam, if not in handbook, question must supply. We can flag. We can use equations from handbook. Need not invent constants? We can present typical constants and say confirm if provided. But exam may provide tables. We can include HDS-5 inlet control constants tables from handbook pages 376-378? The handbook includes constants. We can summarize.

Need craft final. Could be massive. We need fit. Let's plan.

Title: PE Civil: Construction — Stormwater Collection & Drainage Exam Review. Disclaimer. How to use. NCEES spec mapping.

Sections:

1. Core Hydraulic Principles for Stormwater

- Continuity Q=Av
- Energy: z+y+v^2/2g = total head; specific energy E=y+αv^2/2g = y+Q^2/(2gA^2)
- Reynolds, hydraulic radius R=A/P
- Pressure/velocity distributions. Closed conduit as needed: Hazen-Williams, Darcy-Weisbach, minor losses.

2. Open-Channel Flow

- Uniform flow Manning Q=(1.486/n) A R^(2/3) S^(1/2) USCS; SI Q=(1/n) A R^(2/3) S^(1/2). Chezy v=C sqrt(R S).

- Normal depth, critical depth, Froude Fr=v/sqrt(g D) or V/sqrt(g y) rectangular. Critical for rectangular yc= (q^2/g)^(1/3), E_min=1.5 yc. Circular critical charts.
- Specific energy diagram alternate depths; subcritical/supercritical.
- GVF profiles M,S,C,H,A; composite slopes.
- Hydraulic jump: sequent depths rectangular: y2/y1 = 0.5(sqrt(1+8Fr1^2)-1); energy loss ΔE = (y2-y1)^3/(4 y1 y2); efficiency. Momentum equation M=Q^2/(gA)+A ybar.
- Circular pipes partially full: hydraulic elements graph. Q/Qfull and V/Vfull ratios. Max Q at ~0.94D? Max V at ~0.81D? Need correct: For circular pipe, maximum discharge occurs at d/D ~0.94 (Q/Qfull ~1.08), maximum velocity at d/D ~0.81 (V/Vfull ~1.14). Handbook graph. Critical depth charts. We can mention.
- Best hydraulic sections: trapezoid half hexagon etc. From handbook table.

3. Culverts

- Types/materials.
- USGS classification Types 1-6 as handbook. Summarize table.
- Inlet vs outlet control. Headwater HW, tailwater TW. Inlet control if inlet controls; outlet if outlet controls. Compare HW from both.
- Inlet control equations: unsubmerged weir: HW/D = K (Q/(A D^0.5))^M + K_s S? Actually handbook: HW/D = K (Q/(A D^0.5))^M + K_s S? Let's parse. It says:
  Unsubmerged Weir Discharge:
  HW_i/D = K (Q/(A D^0.5))^M + K_s S? Actually image text:
  "HW/D = K (Q/(A D^0.5))^M + K_s S" maybe plus? Let's see handbook page 375:
  Unsubmerged Weir Discharge:
  HW_i/D = K (Q/(A D^0.5))^M + K_s S? It shows:

$$\frac{HW}{D} = K\left(\frac{Q}{AD^{0.5}}\right)^{M} + K_s S$$

or maybe "K Q/(A D^0.5)^M + K_s S"? The text: "HW/D = K (Q/(A D^0.5))^M + K_s S" but OCR:
" /
.
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HW
D
H
K AD
K Q
K S
Q AD
3 5

up to
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u
M
S
0 5
0 5
=
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+
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H"
Mess. Need not overdo; can state HDS-5 form:
Unsubmerged: $HW/D = K(Q/(AD^{0.5}))^M + K_sS$? Actually FHWA HDS-5:

$$\frac{HW}{D} = K\left(\frac{Q}{AD^{0.5}}\right)^M + K_sS$$

where K_s = slope correction (-0.5 for mitered inlets, +0.7? Actually K_s = -0.5 for mitered inlets? Handbook says "KS = slope correction, –0.5 (mitered inlets + 0.7)" confusing.
Submerged orifice:

$$HW/D = c(Q/(AD^{0.5}))^2 + Y + K_sS$$

Maybe c? It shows "c AD K Q Y K S Q AD 4.0 when ...". HDS-5:

$$\frac{HW}{D} = c\left(\frac{Q}{AD^{0.5}}\right)^2 + Y + K_sS$$

Yes.

- Constants K,M,c,Y from charts/tables. We can present some common circular concrete: square edge headwall K=0.0098 M=2.0 c=0.0398 Y=0.67; groove end headwall K=0.0018 M=2.0 c=0.0292 Y=0.74; groove end projecting K=0.0045 M=2.0 c=0.0317 Y=0.69. For CMP headwall K=0.0078 M=2.0 c=0.0379 Y=0.69; mitered K=0.0210 M=1.33 c=0.0463 Y=0.75; projecting K=0.0340 M=1.50 c=0.0553 Y=0.54. Good.
- Outlet control: energy equation; H = He + Hf + Ho + Hb + Hj + Hg. Entrance loss He=ke v^2/2g. Friction Hf = (Ku n^2 L / R_H^(4/3))? Handbook: Hf = [(1 + ke + Ku n^2 L / R_H^(4/3))? Actually page 374:

$$H_f = \left(\frac{1 + k_e + K_u n^2 L}{R_H^{4/3}}\right) \frac{v^2}{2g}?$$

Wait OCR: "H k R K n L g v 1 2 . f e H u 1 33 2 2 = + + f p" Maybe $H_f = \left(\frac{K_u n^2 L}{R_H^{4/3}}\right) \frac{v^2}{2g}$? The text: "Friction Head Loss H k R K n L g v 1 2 . f e H u 1 33 2 2 = + + f p" Could be $H_f = \left(1 + k_e + \frac{K_u n^2 L}{R_H^{4/3}}\right) \frac{v^2}{2g}$? That seems combining entrance? Actually outlet control total head loss H = (1 + ke + (Ku n^2 L)/R_H^{4/3}) V^2/(2g). Let's verify FHWA: $H = \left(1 + k_e + \frac{29 n^2 L}{R^{1.33}}\right) \frac{V^2}{2g}$? Yes for USCS Ku=29? Actually $K_u = 29$ in USCS? In HDS-5: H = (1 + ke + (29 n^2 L)/R^1.33) V^2/(2g) for USCS? Wait Manning in USCS has 1.486, so friction slope S_f = (n V / (1.486 R^{2/3}))^2 = n^2 V^2 /(2.208 R^{4/3}). Head loss = L S_f = n^2 V^2 L /(2.208 R^{4/3}) = (0.453 n^2 L / R^{4/3}) V^2/(2g)? Since 1/2.208 *2g? Hmm. FHWA uses $H = \left(1 + k_e + \frac{29 n^2 L}{R^{1.33}}\right) \frac{V^2}{2g}$? Let's check: 29 n^2 L / R^1.33 times V^2/2g. For n=.012, L=100, R=1, V=10: 29*.000144*100=0.4176; V^2/2g=1.55; Hf=0.65 ft. Manning: S=(n V/(1.486 R^2/3))^2= (.012*10/1.486)^2=.0065; hf=0.65. Yes. So formula H = (1 + ke + 29 n^2 L / R^{1.33}) V^2/(2g)? Actually 29 n^2 L / R^{1.33} = 0.4176, plus 1+ke=1.5, total 1.9176*1.55=2.97 ft. But Hf only 0.65. So total head loss includes exit? H = (1 + ke + 29 n^2 L / R^1.33) V^2/2g. Yes. We'll present.

- Composite roughness $n_c = [\sum(p_i n_i^{1.5})/P]^{0.57}$? Handbook: $n_c = \left[\frac{\sum p_i n_i^{1.5}}{P}\right]^{0.57}$? OCR: "n P p n . . c i i i G 1.5 1 0 57 = =

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  j" Yes exponent 0.57. Actually formula: \(n_c = \left[\frac{\sum (p_i n_i^{1.5})}{P}\right]^{2/3}\)?
  Wait HDS-5: \(n_c = \left[\frac{\sum (p_i n_i^{1.5})}{P}\right]^{2/3}\)? Let's recall: Composite
  Manning's n = [Σ(P_i n_i^{1.5})/P]^{2/3}? The handbook says exponent 0.57, which is ~2/3? No 2/3=0.667.
  0.57 maybe OCR of 2/3? Actually text: "n P p n . . c i i i G 1.5 1 0 57 = =
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  j" could be $n_c = [\sum p_i n_i^{1.5}/P]^{0.57}$. HDS-5 formula is $n_c = [\sum(p_i n_i^{1.5})/P]^{2/3}$. I think 0.57 is OCR of 2/3? Let's use $n_c = [\sum p_i n_i^{1.5}/P]^{2/3}$. But handbook says 0.57? I'll flag use supplied formula.
- Grate loss: $H_g = K_g(W/X)(V_u^2 - V_g^2)/(2g)\sin\theta$? Actually handbook: $H_g = K_g(W/X)\frac{V_u^2 - V_g^2}{2g}\sin\theta$? OCR: "H g V V H K X W g v 1.5 2 2 g g u g g u g 2 2 2 θ = – = f f p p" maybe $H_g = K_g(W/X)\frac{V_u^2 - V_g^2}{2g}\sin\theta$. We'll state.

4. Storm Sewers

- Layout, inlets, manholes, junctions.
- Design storm return period. Rational Q=CiA. Composite C. tc. IDF.
- NRCS curve number Q=(P-0.2S)^2/(P+0.8S), S=1000/CN-10. Initial abstraction.
- Kinematic wave.
- Pipe sizing Manning, partial full flow, min/max velocity. HGL/EGL. Minor losses.
- Inlet capacity: curb, gutter, grate, combination. Curb/gutter triangular: Q = K_u/n S_x^1.67 S_L^0.5 T^2.67? Actually handbook: Q = (K_u/n) S_x^1.67 S_L^0.5 T^2.67; d = T S_x; v = Q/A. Composite gutter sections formulas. We can include.
- Design steps iterative.

5. Watersheds

- Characteristics, IDF, tc methods: sheet flow, shallow concentrated, channel. SCS lag, Kirpich. Hydrographs: unit hydrograph, Snyder, SCS, Clark. Stream gauging. Stormwater management: detention/retention, infiltration, swales. Routing modified Puls. Erosion RUSLE, mulch.

6. Construction Applications & QC

- Temporary drainage/dewatering, ESC plans BMPs, SWPPP, inspection maintenance, spec conformance, common field/design errors.

7. Common errors and exam traps.
8. Worked examples. Need provide multiple. Maybe 5-8 examples covering Manning, critical depth, hydraulic jump, Rational, culvert inlet control, outlet control, gutter, detention routing? We can include detailed solutions. Need practice problems with solutions. Maybe include 10 practice problems with solutions. This will be long. We can be concise but adequate.

Need maybe include tables. We can make tables for Manning n, runoff coefficients, CN, inlet control constants, USGS culvert types, GVF profiles, best hydraulic sections, common errors.

Need end with high-yield checklist and self-test. Self-test maybe 20 questions with answers.

Need ensure no guaranteed pass. Include disclaimer at top and end.

Need maybe mention exam specs. Current NCEES PE Civil: Construction exam specs? The user asked align with current. We can say based on NCEES PE Civil Construction specs and PE Civil Reference Handbook v2.0. If any detail not specified, flag. We can note that exam is CBT, handbook supplied; standards like HDS-5/HEC-22 may not be fully supplied; any required constants will be in question.

Need maybe cite sections from handbook: 6.4, 6.4.10, 6.5, 6.3, 6.2, Chapter 3, Chapter 2.

Need be careful with units. Use USCS primarily. Include SI forms.

Let's craft. We can write in a professional instructor tone. Use headings. Since final answer can be long. We need avoid analysis. Let's produce.

Potential issue: The handbook text huge; we can reference page numbers? Not necessary but mention section.

Let's structure:

- How to use this review
- NCEES exam focus
- 1. Disclaimer

Then sections.

Let's write formulas in LaTeX.

Detailed content:

Open-Channel Flow
Definitions:

- Uniform flow: depth/velocity constant; normal depth yn.
- Critical flow: Fr=1; specific energy minimum; yc.
- Froude: $Fr = V/\sqrt{gD}$ where $D = A/T$. For rectangular, $D = y$, $Fr = V/\sqrt{gy}$.
- Manning: $Q = (1.486/n)AR_H^{2/3}S^{1/2}$ (USCS); $Q = (1/n)AR_H^{2/3}S^{1/2}$ (SI). $R_H = A/P$.
- Chezy: $V = C\sqrt{R_H S}$; $C = (1.486/n)R_H^{1/6}$ USCS.
- Specific energy: $E = y + \alpha V^2/(2g) = y + Q^2/(2gA^2)$. For rectangular, $E = y + q^2/(2gy^2)$.
- Critical depth rectangular: $y_c = (q^2/g)^{1/3}$; $E_min = 3/2y_c$; $V_c = \sqrt{gy_c}$.

- Hydraulic jump: $y_2/y_1 = 0.5(\sqrt{1+8Fr_1^2}-1)$; $\Delta E = (y_2-y_1)^3/(4y_1y_2)$; $Fr_2 = Fr_1(y_1/y_2)^{3/2}$? Actually $Fr_2 = Fr_1(y_1/y_2)^{3/2}$. Energy loss also $\Delta E = (y_2-y_1)^3/(4y_1y_2)$. Efficiency.
- GVF profiles table: Mild M1, M2, M3; Steep S1,S2,S3; Critical C1,C2; Horizontal H2,H3; Adverse A2,A3. Need correct: H2, H3; A2, A3. M1 backwater above normal; M2 between normal and critical; M3 below critical. S1 above normal/critical? Actually steep: yn<yc. S1: y>yc>yn? Wait steep slope normal depth < critical. S1: actual > yc > yn? subcritical? Let's recall: S1: y > yc > yn? Actually on steep slope, yn<yc. Zone 1 above yc and yn: y>yc>yn, subcritical, backwater curve. S2: between yc and yn? But yn<yc, so between yn and yc: supercritical? S2 is drawdown. S3: below both, supercritical. We'll table.
- Composite slopes: transitions between mild/steep, hydraulic jumps/drops, control at critical depth.

Culverts

USGS types table from handbook:

Type 1: steep, partial, inlet not submerged, outlet not submerged, critical depth at inlet, inlet control. Eq Q=CAc sqrt(2g(h1 - z + ...))? We can summarize.

Type 2: mild, partial, critical depth at outlet, outlet control? Actually Type 2 critical depth at outlet, outlet control.

Type 3: tranquil flow throughout, mild, partial, no submerged, outlet? It's outlet control? Type 3 equation.

Type 4: submerged outlet, any slope, full, yes yes, outlet control.

Type 5: rapid flow at inlet, any, partial, yes no, inlet control.

Type 6: full flow free outfall, any, full, yes no, inlet control.

We can table.

Inlet control design:

- Unsubmerged: $HW/D = K(Q/(AD^{0.5}))^M + K_sS$.
- Submerged: $HW/D = c(Q/(AD^{0.5}))^2 + Y + K_sS$.
- Constants table common circular. Use HDS-5 Chart 1,2 etc.
  Outlet control:
- $H = H_e + H_f + H_o + H_b + H_j + H_g$.
- $H_e = k_eV^2/(2g)$.
- $H_f = \frac{K_un^2L}{R_H^{4/3}}\frac{V^2}{2g}$ with Ku=29 USCS? Then total $H = \left(1 + k_e + \frac{29n^2L}{R_H^{4/3}}\right)\frac{V^2}{2g}$ for full flow? Actually includes exit loss 1.0. Yes.
- Outlet control headwater $HW = H + TW - LS$? Actually $HW = H + h_o - LS_0$? From energy: $HW = H + TW - LS$? Let's derive: For full flow, $H = (1 + k_e + ...)V^2/2g$. HGL at outlet = TW. Upstream HW = H + TW - LS? Actually energy equation: $HW + z_1 + V_u^2/2g = TW + z_2 + V_d^2/2g + H_L$. If invert at inlet z1, outlet z2 = z1 - LS. Ignore approach/downstream velocity. Then $HW + z1 = TW + z2 + H_L = TW + z1 - LS + H_L$. So $HW = TW - LS + H_L$. Yes.
- Composite roughness.

Storm Sewers

Rational Q=CiA (Q cfs, i in/hr, A acres). Composite C = ΣCiAi/ΣAi. tc. IDF. NRCS. Kinematic wave. Manning. Min velocities: 2 ft/s? Typical storm sewer min 2 ft/s? Max 10-15 ft/s? Flag local. HGL/EGL. Minor losses. Inlet capacity.

Curb/gutter triangular:
$Q = (K_u/n)S_x^{1.67}S_L^{0.5}T^{2.67}$, $K_u = 0.56$ USCS. $d = TS_x$. $v = Q/A$, $A = Td/2 = T^2S_x/2$. Composite: $Q = Q_1 + Q_2$ with sections. Spread T.

Watersheds

- IDF: i = c T^e / (T_d + f)^m? Handbook Eq 1-3. We can summarize.
- tc: sheet flow $T_t = 0.007(nL)^{0.8}/(P_2^{0.5}S^{0.4})$? Actually HEC-22: $T_t = 0.007(nL)^{0.8}/(P_2^{0.5}S^{0.4})$ where P2 2-yr 24-hr rainfall? In handbook: Sheet flow formula with Ku=0.933 USCS? Let's check page 404: $T_{ti} = \frac{K_u(nL)^{0.8}}{(P_2)^{0.5}S^{0.4}}$? OCR: "Tti = sheet flow travel time; n roughness; L flow length; I rainfall intensity; S slope; Ku empirical coefficient equal to 6.92 (0.933 in USCS units)" Hmm. HEC-22: $T_t = \frac{0.007(nL)^{0.8}}{(P_2)^{0.5}S^{0.4}}$ where P2 = 2-yr, 24-hr rainfall (in). If I is intensity, Ku=0.933? We can state use HEC-22 formula if supplied.
- SCS lag: $t_L = 0.000526[1000/CN - 9]^{0.7}L^{0.8}/S^{0.5}$? Handbook: $t_L = 0.000526(1000/CN - 9)^{0.7}L^{0.8}S^{0.5}$? Actually page 391: $t_L = 0.000526\left(\frac{1000}{CN} - 9\right)^{0.7}L^{0.8}S^{0.5}$? OCR: "t 0.000526 1,000 9 L CN S . . . L 0.8 0.7 0.5 = – – d n" It has $S^{0.5}$ in denominator? Hmm. HEC: $t_L = \frac{0.000526(1000/CN-9)^{0.7}L^{0.8}}{S^{0.5}}$? Actually SCS lag: $t_L = \frac{0.000526(1000/CN-9)^{0.7}L^{0.8}}{S^{0.5}}$? I think yes. We'll write.
- Kirpich: $t_c = 0.0078L^{0.77}S^{-0.385}$ (min, L ft, S ft/ft). Actually Kirpich: $t_c = 0.0078L^{0.77}S^{-0.385}$.
- Unit hydrograph: SCS triangular $q_p = 484AQ/t_p$; $t_p = D/2 + t_L$; $T_b = 2.67t_p$. Snyder: $t_p = C_t(LL_c)^{0.3}$; $q_p = 640C_pA/t_p$; $W_{50} = 2.5q_p^{-1.08}$? We can summarize.
- Detention routing: Modified Puls: $(I_1 + I_2)/2 + (S_1/\Delta t - O_1/2) = S_2/\Delta t + O_2/2$. Storage volume V_s = max(V_in - V_out). RUSLE A=R K L S C P.

Construction Applications

- Temporary drainage, dewatering, ESC, SWPPP, inspections.

Common errors as listed.

Worked examples:
Let's create 6 examples with detailed solutions.

Example 1: Manning normal depth. A trapezoidal channel b=6 ft, z=2, n=0.025, S=0.001, Q=120 cfs. Find normal depth. Use iterative. Show solve.
Trial y=2.5: A= (b+zy)y=(6+5)*2.5=27.5; P=b+2y sqrt(1+z^2)=6+2*2.5*sqrt5=6+11.18=17.18; R=1.60; Q=(1.486/.025)27.51.60^(2/3)0.001^0.5 =59.44*27.5*1.370.03162=69.2? Too low. Need y~3.5. y=3.5: A=(6+7)3.5=45.5; P=6+2*3.5*2.236=6+15.65=21.65; R=2.10; R^(2/3)=1.64; Q=59.44*45.5*1.64.03162=140.4. y=3.2: A=(6+6.4)3.2=39.68; P=6+2*3.2*2.236=20.31; R=1.954; R^(2/3)=1.56; Q=59.44*39.68*1.56.03162=116.5. y=3.23: A=(6+6.46)3.23=40.24; P=6+14.44=20.44; R=1.969; R^(2/3)=1.57;

Q=59.44*40.24*1.57.*03162=118.3. y=3.25: A=(6+6.5)*3.25=40.625; P=6+14.53=20.53; R=1.979; R^(2/3)=1.576; Q=59.44*40.625*1.576*.03162=119.9. So yn≈3.25 ft.

Example 2: Critical depth rectangular. Q=200 cfs, b=10 ft. q=20 cfs/ft. yc=(q^2/g)^(1/3)= (400/32.2)^(1/3)=12.42^(1/3)=2.32 ft. E_min=3.48 ft. Fr at y=3 ft? V=Q/A=200/(10*3)=6.67; Fr=6.67/sqrt(32.2*3)=6.67/9.83=0.68 subcritical. At y=2 ft: V=10; Fr=10/sqrt(64.4)=1.25 supercritical.

Example 3: Hydraulic jump. Rectangular channel, y1=1.0 ft, V1=20 ft/s. Fr1=20/sqrt(32.2*1)=3.52. y2=0.5(sqrt(1+8*12.4)-1)=0.5(sqrt(100.2)-1)=0.5(10.01-1)=4.50 ft. Energy loss ΔE=(4.5-1)^3/(4*1*4.5)=42.875/18=2.38 ft. Check.

Example 4: Rational method. Area 25 acres, C=0.45, tc=20 min, 10-yr IDF i=4.2 in/hr. Q=0.45*4.2*25=47.25 cfs. If composite C with 15 acres C=.3, 10 acres C=.7: C=(15*.3+10*.7)/25=(4.5+7)/25=0.46.

Example 5: Culvert inlet control. Circular concrete pipe D=3 ft, Q=120 cfs, square edge with headwall. K=0.0098, M=2.0, c=0.0398, Y=0.67. A=πD^2/4=7.069 ft^2. Unsubmerged: HW/D = 0.0098*(120/(7.069*sqrt(3)))^2 + 0? Actually M=2.0. Q/(A D^0.5)=120/(7.069*1.732)=120/12.24=9.80. Square=96.0. *0.0098=0.941. So HW/D=0.941, HW=2.82 ft. Submerged: HW/D = c*(Q/(A D^0.5))^2 + Y =0.0398*96.0+0.67=3.82+0.67=4.49, HW=13.47. Inlet control uses larger? Actually inlet control equation: compare unsubmerged and submerged; use larger? For HW/D > 4? Wait unsubmerged applies up to HW/D~? Submerged when HW/D > 4.0? The equations are separate; if HW/D > 4 use submerged. Here unsubmerged gives 0.941, submerged gives 4.49; actual likely unsubmerged? But inlet control nomograph gives HW/D ~2.1? Let's check. For Q=120, D=3, Q/D^2.5? HDS chart. Maybe my K constant wrong? Square edge headwall K=0.0098, M=2.0, c=0.0398, Y=0.67. Q/(A D^0.5) = 120/(7.069*1.732)=9.80; squared=96.0; unsub=0.0098*96=0.94. HW=2.82 ft. That seems plausible. Submerged equation not applicable until HW/D>4? The text says submerged when HW/D > 4.0? Actually "Submerged Orifice Discharge: HW/D = c(...)^2 + Y + K_s S when HW/D > 4.0"? So use unsub if result <4. So HW=2.82 ft. Good.

Example 6: Outlet control. Pipe D=3 ft, L=200 ft, n=0.012, ke=0.5, Q=120 cfs, TW=3 ft, slope S=0.001. Full flow area A=7.069, V=Q/A=16.98 ft/s. R=D/4=0.75. H = (1+ke + 29 n^2 L / R^1.33) V^2/(2g). 29.*000144*200=0.8352; R^1.33=0.75^1.33=0.682; term=1.225; total=1+0.5+1.225=2.725; V^2/2g=16.98^2/64.4=4.48; H=12.21 ft. HW=H+TW-LS=12.21+3-0.2=15.01 ft. Compare inlet HW=2.82, outlet controls, HW=15.0 ft.

Example 7: Gutter flow. Triangular gutter Sx=0.02, SL=0.01, n=0.016, T=8 ft. Q=(0.56/0.016)*(0.02^1.67)* (0.01^0.5)*(8^2.67). Compute 0.56/0.016=35; 0.02^1.67=0.00153? 0.02^1.67 = exp(1.67 ln .02)=exp(1.67-3.912)=exp(-6.53)=0.00146. 0.01^0.5=0.1. 8^2.67=exp(2.67*2.079)=exp(5.55)=257. Q=35*0.00146*0.1*257=1.31 cfs. d=T*Sx=0.16 ft. A=T*d/2=8*0.16/2=0.64 ft^2. V=Q/A=2.05 ft/s.

Example 8: NRCS runoff. P=4 in, CN=80. S=1000/80-10=12.5-10=2.5 in. Q=(4-0.2*2.5)^2/(4+0.8*2.5)=(4-0.5)^2/(4+2)=12.25/6=2.04 in.

Example 9: Detention routing simple? Maybe too much. We can include.

Practice problems with solutions: Could be shorter. Maybe 10 problems. But we already have examples. User requested practice problems with detailed solutions. We can integrate examples as practice problems. At end,

> self-test with answers only.
>
> Need end with high-yield checklist and self-test. Self-test: 20 questions, answers.
>
> Need mention "flag if not specified". For constants, say use provided HDS-5 tables; if not supplied, question will provide.
>
> Now write final. It will be long. Need ensure formatting. Use markdown.
>
> Let's draft. We need not include every table from handbook, but enough. We'll include tables with typical values.
>
> Potential issue: The final answer may exceed desired length. But task demands comprehensive. We'll be thorough but not insane.
>
> Let's start final.

PE Civil: Construction — Stormwater Collection & Drainage

Exam-Focused Review for the NCEES PE Civil: Construction Exam

Based on NCEES PE Civil Construction exam specifications and the *PE Civil Reference Handbook*, Version 2.0.
Scope: Only stormwater collection and drainage within Hydraulics and Hydrology. Related closed-conduit, hydraulics, geotechnical, and construction topics are included only as they directly support stormwater collection and drainage.

> **Disclaimer:** This review is exam-preparation material designed to maximize readiness. It does not guarantee a passing score. Always follow the current NCEES exam specifications and the supplied *PE Civil Reference Handbook* during the exam. If a constant, chart, or local criterion is not supplied by NCEES or in the exam question, do not invent it—flag it and use only what is provided.

0. How to Use This Review

Priority	Topic	Why It Matters
1	Manning’s equation, normal/critical depth, Froude number	Backbone of open-channel and storm sewer design

Priority	Topic	Why It Matters
2	Culvert inlet vs. outlet control	Very common PE problems
3	Rational Method + time of concentration	Peak flow for storm sewers
4	NRCS/SCS runoff + unit hydrograph	Watershed and detention problems
5	Curb/gutter/inlet capacity	Construction/transportation drainage
6	Detention/retention routing	Stormwater management
7	Common errors and unit traps	Prevents missed “easy” points

Exam strategy:

- Write units on every line.
- Check whether flow is pressure/full, partially full, or open-channel.
- For culverts, always compute both inlet-control and outlet-control headwater; the higher controls.
- For storm sewers, check HGL, not just pipe size.
- For Rational Method, ensure i corresponds to t_c and the correct return period.
- For NRCS, ensure CN matches land use and soil group.

1. Core Hydraulic Principles for Stormwater

1.1 Continuity and Energy

Continuity:

$$Q = AV$$

$$A_1V_1 = A_2V_2$$

Energy equation (Bernoulli with losses):

$$z_1 + y_1 + \frac{\alpha_1 V_1^2}{2g} = z_2 + y_2 + \frac{\alpha_2 V_2^2}{2g} + h_f$$

where:

- z = elevation of channel bottom or pipe invert
- y = depth of flow
- V = mean velocity
- α = kinetic energy correction factor, usually 1.0
- h_f = head loss

Total head:

$$H = z + y + \frac{V^2}{2g}$$

Specific energy:

$$E = y + \frac{\alpha V^2}{2g} = y + \frac{Q^2}{2gA^2}$$

Hydraulic radius:

$$R_H = \frac{A}{P}$$

where A = cross-sectional flow area, P = wetted perimeter.

Reynolds number (open channel):

$$Re = \frac{V R_H}{\nu}$$

- Laminar: $Re < 500$
- Transitional: $500 < Re < 2{,}000$
- Turbulent: $Re \geq 2{,}000$

Reynolds number (circular pipe):

$$Re = \frac{VD}{\nu}$$

- Laminar: $Re < 2{,}000$
- Transitional: $2{,}000 < Re < 4{,}000$
- Turbulent: $Re \geq 4{,}000$

1.2 Closed Conduit Flow for Storm Sewers

Hazen–Williams

$$V = k_1 C R_H^{0.63} S^{0.54}$$

$$Q = k_1 C A R_H^{0.63} S^{0.54}$$

- USCS: $k_1 = 1.318$
- SI: $k_1 = 0.849$

Circular pipe, USCS:

$$Q = 0.432 C D^{2.63} S^{0.54}$$

$$h_f = \frac{4.73 L Q^{1.852}}{C^{1.852} D^{4.87}}$$

where Q in cfs, D in ft, L in ft.

Darcy–Weisbach

$$h_f = f \frac{L}{D} \frac{V^2}{2g}$$

- $f = f(Re, \varepsilon/D)$ from Moody diagram.
- Use for storm sewers when f or roughness is supplied.

Minor Losses

$$h_m = K\frac{V^2}{2g}$$

Typical K:

- Sharp entrance: 0.5
- Round entrance: 0.1
- Projecting pipe entrance: 0.8
- Exit: 1.0
- Globe valve, fully open: 10.0
- Gate valve, fully open: 0.2
- 90° short-radius elbow: 0.9
- 90° long-radius elbow: 0.6
- 45° elbow: 0.4
- Tee, flow through run: 0.6
- Tee, flow through branch: 1.8

2. Open-Channel Flow

2.1 Uniform Flow and Manning's Equation

Manning's equation:

$$Q = \frac{1.486}{n} A R_H^{2/3} S^{1/2} \quad \text{(USCS)}$$

$$Q = \frac{1}{n} A R_H^{2/3} S^{1/2} \quad \text{(SI)}$$

$$V = \frac{1.486}{n} R_H^{2/3} S^{1/2} \quad \text{(USCS)}$$

Conveyance:

$$K = \frac{1.486}{n} A R_H^{2/3}$$

$$Q = K S^{1/2}$$

Chezy equation:

$$V = C\sqrt{R_H S}$$

$$C = \frac{1.486}{n} R_H^{1/6} \quad \text{(USCS)}$$

Manning's n — Typical Values

Material	n
Brass, copper	0.010–0.011
Welded steel	0.012
Riveted steel	0.016
Cast iron, coated	0.013
Corrugated metal storm drain	0.024
Concrete culvert	0.013
Concrete-lined channel	0.015
Clay	0.013
Brickwork	0.013
PVC	0.009–0.011
Earth channel, clean	0.022
Earth channel, winding	0.025

Material	n
Rock cut	0.035
Natural stream, clean straight	0.030
Natural stream, winding, pools	0.040
Natural stream, sluggish, weedy	0.070

2.2 Normal Depth, Critical Depth, Froude Number

Normal depth y_n: depth for uniform flow; solve Manning implicitly.

Critical depth y_c: depth at minimum specific energy; $Fr = 1$.

Froude number:

$$Fr = \frac{V}{\sqrt{gD}}$$

where:

$$D = \frac{A}{T}$$

T = top width.

For a rectangular channel:

$$D = y$$

$$Fr = \frac{V}{\sqrt{gy}}$$

Flow classification:

- $Fr < 1$: subcritical, tranquil
- $Fr = 1$: critical

- $Fr > 1$: supercritical, rapid

Critical depth — rectangular channel:

$$y_c = \left(\frac{q^2}{g}\right)^{1/3}$$

where $q = Q/b$.

Critical velocity — rectangular:

$$V_c = \sqrt{g y_c}$$

Minimum specific energy — rectangular:

$$E_{\min} = \frac{3}{2} y_c$$

Critical flow general:

$$\frac{Q^2 T}{g A^3} = 1$$

2.3 Specific Energy and Alternate Depths

Specific energy diagram:

$$E = y + \frac{V^2}{2g}$$

- For a given $E > E_{\min}$, two possible depths: **alternate depths**.
- Upper limb: subcritical.
- Lower limb: supercritical.
- At $E_{\min}$: critical depth.

Rectangular channel:

$$E = y + \frac{q^2}{2gy^2}$$

2.4 Gradually Varied Flow (GVF) Profiles

Slope classification:

Type	Name	Condition
M	Mild	$y_n > y_c$
S	Steep	$y_n < y_c$
C	Critical	$y_n = y_c$
H	Horizontal	$S_0 = 0$
A	Adverse	$S_0 < 0$

Profile zones:

- Zone 1: actual depth > y_c and y_n — subcritical
- Zone 2: actual depth between y_c and y_n — can be subcritical or supercritical
- Zone 3: actual depth < both — supercritical

Common profiles:

- M1: backwater curve above normal depth
- M2: drawdown between normal and critical
- M3: supercritical below critical
- S1: backwater above critical and normal
- S2: drawdown between critical and normal
- S3: supercritical below both
- C1, C2: critical slope profiles
- H2, H3: horizontal slope

- A2, A3: adverse slope

Composite slopes: Transitions between mild/steep slopes create controls, hydraulic jumps, or drawdown. Always locate critical depth at slope breaks or inlet/outlet controls.

2.5 Rapidly Varied Flow and Hydraulic Jump

Momentum equation:

$$M = \frac{Q^2}{gA} + A\bar{y}$$

where $\bar{y}$ = distance from water surface to centroid of area.

Sequent depths — rectangular channel:

$$\frac{y_2}{y_1} = \frac{1}{2}\left(\sqrt{1 + 8Fr_1^2} - 1\right)$$

$$\frac{y_1}{y_2} = \frac{1}{2}\left(\sqrt{1 + 8Fr_2^2} - 1\right)$$

Energy loss:

$$\Delta E = \frac{(y_2 - y_1)^3}{4y_1 y_2}$$

Jump classification:

Fr_1	Type
1.0–1.7	Undular
1.7–2.5	Weak
2.5–4.5	Oscillating
4.5–9.0	Steady

Fr_1	Type
> 9.0	Strong

2.6 Flow in Circular Pipes Partially Full

Hydraulic elements graph gives ratios:

- Q/Q_{full}
- V/V_{full}
- A/A_{full}
- R/R_{full}
- d/D

Key facts:

- Maximum discharge in a circular pipe occurs at $d/D \approx 0.94$, where $Q/Q_{\text{full}} \approx 1.08$.
- Maximum velocity occurs at $d/D \approx 0.81$, where $V/V_{\text{full}} \approx 1.14$.
- At $d/D = 0.5$, $Q/Q_{\text{full}} \approx 0.5$, $V/V_{\text{full}} \approx 1.0$.

Critical depth charts are used for circular pipes partially full.

Minimum diameter under full flow:

$$Q = \frac{C_0}{n} D^{8/3} S^{1/2}$$

where $C_0 = 2.16$ for USCS.

2.7 Best Hydraulic Sections

Section	Area A	Wetted Perimeter P	Hydraulic Radius R	Top Width T
Trapezoid, half hexagon	$1.5y^2\sqrt{3}$	$2y\sqrt{3}$	$y/2$	$2y\sqrt{3}$

Section	Area A	Wetted Perimeter P	Hydraulic Radius R	Top Width T
Rectangle, half square	$2y^2$	$4y$	$y/2$	$2y$
Triangle, half square	y^2	$2y\sqrt{2}$	$y/(2\sqrt{2})$	$2y$
Semicircle	$\frac{\pi}{2}y^2$	πy	$y/2$	$2y$
Parabola	$\frac{4}{3}y^2$	$2.295y$	$0.58y$	$2.83y$

Best hydraulic section: maximizes R_H for a given area, thus maximizes Q.

3. Culverts

3.1 Culvert Types and Materials

- **Shapes:** circular, box, arch, elliptical, pipe-arch.
- **Materials:** concrete, corrugated metal (CMP), plastic, aluminum, steel.
- **Inlets:** projecting, headwall, wingwall, mitered, beveled, tapered.

3.2 USGS Culvert Flow Classification

USGS Type	Typical Slope	Flow Type	Submerged Inlet?	Submerged Outlet?	Control Section
1	Steep	Partially full	No	No	Inlet
2	Mild	Partially full	No	No	Outlet
3	Mild	Tranquil throughout	No	No	Outlet
4	Any	Full	Yes	Yes	Outlet
5	Any	Partial	Yes	No	Inlet

USGS Type	Typical Slope	Flow Type	Submerged Inlet?	Submerged Outlet?	Control Section
6	Any	Full free outfall	Yes	No	Inlet

Type 1: critical depth at inlet, inlet control.
Type 2: critical depth at outlet, outlet control.
Type 3: tranquil flow throughout, outlet control.
Type 4: submerged outlet, full flow, outlet control.
Type 5: rapid flow at inlet, inlet control.
Type 6: full flow, free outfall, inlet control.

3.3 Inlet vs. Outlet Control

Inlet control: inlet geometry controls capacity; barrel flows partially full or just full at inlet.
Outlet control: outlet/tailwater/barrel friction controls; barrel may flow full or partially full.

Procedure:

1. Assume inlet control. Compute HW_i.
2. Assume outlet control. Compute HW_o.
3. Actual headwater $HW = \max(HW_i, HW_o)$.
4. If $HW_i > HW_o$, inlet controls. If $HW_o > HW_i$, outlet controls.

3.4 Inlet Control Equations (FHWA HDS-5)

Unsubmerged (weir):

$$\frac{HW}{D} = K\left(\frac{Q}{AD^{0.5}}\right)^{M} + K_s S$$

Submerged (orifice):

$$\frac{HW}{D} = c\left(\frac{Q}{AD^{0.5}}\right)^{2} + Y + K_s S$$

where:

- Q = discharge
- A = full barrel area
- D = interior height
- S = barrel slope
- K, M, c, Y = constants from HDS-5 tables
- K_s = slope correction: -0.5 for mitered inlets, $+0.7$ for others

Common circular concrete constants:

Inlet Configuration	K	M	c	Y
Square edge w/ headwall	0.0098	2.0	0.0398	0.67
Groove end w/ headwall	0.0018	2.0	0.0292	0.74
Groove end projecting	0.0045	2.0	0.0317	0.69
CMP headwall	0.0078	2.0	0.0379	0.69
CMP mitered to slope	0.0210	1.33	0.0463	0.75
CMP projecting	0.0340	1.50	0.0553	0.54

> **Exam tip:** If the question supplies HDS-5 charts, use them. If not, the constants must be given.
> Do not memorize all charts—know how to read them.

3.5 Outlet Control

Total head loss:

$$H = H_e + H_f + H_o + H_b + H_j + H_g$$

where:

- H_e = entrance loss
- H_f = friction loss
- H_o = exit loss
- H_b = bend loss
- H_j = junction loss
- H_g = grate loss

Entrance loss:

$$H_e = k_e \frac{V^2}{2g}$$

Friction loss:

$$H_f = \frac{K_u n^2 L}{R_H^{4/3}} \frac{V^2}{2g}$$

- USCS: $K_u = 29$
- SI: $K_u = 19.6$

Full-flow outlet control (including exit loss):

$$H = \left(1 + k_e + \frac{29 n^2 L}{R_H^{4/3}}\right) \frac{V^2}{2g} \quad \text{(USCS)}$$

Headwater:

$$HW = H + TW - LS_0$$

where TW = tailwater depth above outlet invert, L = culvert length, S_0 = barrel slope.

Entrance loss coefficients k_e:

Structure / Entrance	k_e
Concrete pipe, projecting socket	0.2
Concrete pipe, projecting square cut	0.5

Structure / Entrance	k_e
Concrete pipe, headwall, socket	0.2
Concrete pipe, headwall, square edge	0.5
Concrete pipe, headwall, rounded	0.2
Concrete pipe, mitered to slope	0.7
Concrete pipe, end section conforming	0.5
Concrete pipe, beveled 33.7° or 45°	0.2
CMP, projecting no headwall	0.9
CMP, headwall square edge	0.5
CMP, mitered to slope	0.7
CMP, end section conforming	0.5
CMP, beveled	0.2
Box, headwall parallel, square edge	0.5
Box, headwall parallel, rounded/beveled	0.2
Box, wingwalls 30°–75°, square edge	0.4
Box, wingwalls 30°–75°, rounded/beveled	0.2
Box, wingwalls 10°–25°, square edge	0.5
Box, wingwalls parallel, square edge	0.7
Box, side- or slope-tapered inlet	0.2

3.6 Composite Roughness

$$n_c = \left[\frac{\sum (p_i n_i^{1.5})}{P} \right]^{2/3}$$

where:

- p_i = wetted perimeter influenced by material i
- n_i = Manning's n for material i
- P = total wetted perimeter

> **Flag:** The handbook shows an exponent that may appear as 0.57 in some OCR versions. Use the formula supplied in the exam.

3.7 Grate (Bar Rack) Head Loss

$$H_g = K_g \left(\frac{W}{X} \right) \frac{V_u^2 - V_g^2}{2g} \sin \theta_g$$

where:

- K_g = bar shape factor
  - 2.42 = sharp-edged rectangular
  - 1.83 = rectangular with semicircular upstream face
  - 1.79 = circular bars
  - 1.67 = rectangular with semicircular upstream and downstream faces
- W = max cross-sectional width of bars facing flow
- X = minimum clear spacing between bars
- θ_g = angle of grate with horizontal
- V_u = upstream velocity
- V_g = velocity through grate

4. Storm Sewers

4.1 Storm Sewer System Layout

Components:

- Inlets: curb, gutter, grate, combination
- Manholes
- Junctions
- Pipes
- Outfalls
- Detention/retention facilities

Design considerations:

- Design storm return period (e.g., 2-yr, 10-yr, 25-yr, 100-yr)
- Minimum pipe size (often 12–15 in.)
- Minimum velocity (2 ft/s)
- Maximum velocity (10–15 ft/s)
- Minimum cover
- HGL below gutter line for design storm

4.2 Rational Method

$$Q = CiA$$

where:

- Q = peak discharge (cfs)
- C = runoff coefficient
- i = rainfall intensity (in./hr) for duration t_c
- A = drainage area (acres)

Composite C:

$$C_w = \frac{\sum C_i A_i}{\sum A_i}$$

Runoff coefficients:

Land Use	C
Downtown business	0.70–0.95
Neighborhood business	0.50–0.70
Single-family residential	0.30–0.50
Multi-family detached	0.40–0.60
Multi-family attached	0.60–0.75
Suburban	0.25–0.40
Apartment	0.50–0.70
Light industrial	0.50–0.80
Heavy industrial	0.60–0.90
Parks, cemeteries	0.10–0.25
Playgrounds	0.20–0.40
Railroad yards	0.20–0.40
Unimproved areas	0.10–0.30
Lawns, sandy soil, flat	0.05–0.10
Lawns, sandy soil, average	0.10–0.15
Lawns, sandy soil, steep	0.15–0.20
Lawns, heavy soil, flat	0.13–0.17

Land Use	C
Lawns, heavy soil, average	0.18–0.22
Lawns, heavy soil, steep	0.25–0.35
Asphaltic streets	0.70–0.95
Concrete streets	0.80–0.95
Brick streets	0.70–0.85
Drives and walks	0.75–0.85
Roofs	0.75–0.95

4.3 NRCS (SCS) Rainfall-Runoff Method

$$Q = \frac{(P - 0.2S)^2}{P + 0.8S}$$

$$S = \frac{1000}{CN} - 10$$

where:

- Q = runoff depth (in.)
- P = precipitation (in.)
- S = maximum basin retention (in.)
- CN = curve number

Initial abstraction:

$$I_a = 0.2S$$

Loss:

$$\text{Loss} = P - Q$$

SCS hydrologic soil groups:

- A: deep sand, deep loess, aggregated silts — high infiltration
- B: shallow loess, sandy loam
- C: clay loams, shallow sandy loam, low organic content, high clay
- D: swelling clays, heavy plastic clays, saline soils — low infiltration

Typical CN values:

Land Use	A	B	C	D
Paved parking, roofs	98	98	98	98
Streets, paved w/ curbs	98	98	98	98
Gravel streets	76	85	89	91
Dirt streets	72	82	87	89
Commercial/business	89	92	94	95
Industrial	81	88	91	93
Residential 1/8 ac	77	85	90	92
Residential 1/4 ac	61	75	83	87
Residential 1/3 ac	57	72	81	86
Residential 1/2 ac	54	70	80	85
Residential 1 ac	51	68	79	84
Residential 2 ac	46	65	77	82
Newly graded area	77	86	91	94
Pasture, poor	68	79	86	89
Pasture, fair	49	69	79	84

Land Use	A	B	C	D
Pasture, good	39	61	74	80
Woods, poor	45	66	77	83
Woods, fair	36	60	73	79
Woods, good	25	55	70	77

4.4 Kinematic Wave Runoff Method

$$Q = \alpha A^m$$

$$\alpha = \frac{1.486}{n} S_0^{1/2} P_c^{2/3}$$

where:

- $m = 5/3$ for wide rectangular/trapezoidal
- $m = 1.0$ for circular pipe
- $m = 1.33$ for triangular
- P_c = channel parameter (width, diameter, etc.)

Time to equilibrium:

$$t_c = \frac{nL}{1.49 S_0^{1/2} h^{2/3}}$$

4.5 Pipe Sizing and Slope

Use Manning's equation. For partially full circular pipes, use hydraulic elements graph.

Minimum and maximum velocities:

- Minimum: 2 ft/s (self-cleansing)
- Maximum: 10–15 ft/s (erosion control; local criteria may differ)

HGL and EGL:

- EGL = HGL + $V^2/2g$
- HGL must remain below gutter line for design storm.
- Check for submerged outlet, tailwater, and minor losses.

Minor losses in storm sewers:

- Entrance/exit
- Bends
- Junctions
- Transitions
- Grate inlets

4.6 Inlet Capacity

Curb and Gutter as Triangular Channels

$$Q = \frac{K_u}{n} S_x^{1.67} S_L^{0.5} T^{2.67}$$

$$d = T S_x$$

$$v = \frac{Q}{A}$$

$$A = \frac{T d}{2} = \frac{T^2 S_x}{2}$$

where:

- $K_u = 0.56$ (USCS)
- S_x = cross slope

- S_L = longitudinal slope
- T = spread
- d = depth at curb

Composite Gutter Sections

$$Q = Q_1 + Q_2$$

$$Q_1 = \frac{0.56}{n} S_x^{1.67} S_L^{0.5} T_1^{2.67}$$

$$Q_2 = \frac{0.56}{n} S_x^{1.67} S_L^{0.5} (T^{2.67} - T_1^{2.67})$$

$$d = T S_x$$

Inlet Types

- Curb inlets
- Gutter inlets
- Grate inlets
- Combination inlets

Capacity depends on:

- Spread
- Depth
- Slope
- Grate type
- Clogging factor
- Bypass flow

4.7 Storm Sewer Design Steps

1. Delineate drainage areas.
2. Compute t_c for each area.
3. Determine design storm and IDF intensity.
4. Compute peak flow using Rational Method or NRCS.
5. Layout sewer system.
6. Size pipes using Manning's equation.
7. Check velocities, HGL, and cover.
8. Check inlet capacity.
9. Design detention/retention if required.
10. Prepare ESC/SWPPP.

5. Watersheds

5.1 Watershed Characteristics

- Area
- Shape
- Slope
- Land use
- Soil type
- Time of concentration
- Imperviousness
- Storage

5.2 Rainfall Intensity–Duration–Frequency (IDF)

General forms:

$$i = \frac{cT^e}{T_d^m}$$

$$i = \frac{cT^e}{(T_d + f)^m}$$

where:

- i = intensity (in./hr)
- T_d = duration (min)
- T = return period (years)
- c, e, f, m = coefficients

Procedure:

1. Determine t_c.
2. Use $T_d = t_c$.
3. Read i from IDF curve for design return period.
4. Use i in Rational Method.

5.3 Time of Concentration

Sheet Flow

$$T_t = \frac{K_u (nL)^{0.8}}{(P_2)^{0.5} S^{0.4}}$$

where:

- $K_u = 0.933$ (USCS)
- n = Manning's roughness for overland flow
- L = flow length (ft)
- P_2 = 2-yr, 24-hr rainfall (in.)
- S = slope (ft/ft)

Shallow Concentrated Flow

Use velocity versus slope graph or Manning's equation.

Channel Flow

Use Manning's equation and travel time:

$$T_t = \frac{L}{V}$$

SCS Lag Formula

$$t_L = \frac{0.000526(1000/CN - 9)^{0.7} L^{0.8}}{S^{0.5}}$$

$$t_c = \frac{5}{3} t_L$$

where:

- t_L = lag time (hr)
- L = watershed length (ft)
- S = watershed slope (%)

Kirpich Formula

$$t_c = 0.0078 L^{0.77} S^{-0.385}$$

where:

- t_c = min
- L = ft
- S = ft/ft

5.4 Hydrograph Development

Unit Hydrograph

- 1 unit of runoff (1 in. or 1 mm) uniformly over watershed in specified duration.
- Direct runoff hydrograph.

SCS Triangular Unit Hydrograph:

$$q_p = \frac{484AQ}{t_p}$$

$$t_p = \frac{D}{2} + t_L$$

$$T_b = 2.67t_p$$

where:

- q_p = peak flow (cfs)
- A = area (mi²)
- Q = runoff depth (in.)
- t_p = time to peak (hr)
- D = duration of excess rainfall (hr)
- t_L = lag time (hr)
- T_b = time base (hr)

Snyder Synthetic Unit Hydrograph:

$$t_p = C_t(LL_c)^{0.3}$$

$$q_p = \frac{640C_pA}{t_p}$$

$$W_{50} = 2.5q_p^{-1.08}$$

$$W_{75} = 1.22q_p^{-1.08}$$

where:

- L = stream length from outlet to upstream limits (mi)

- L_c = stream length from outlet to centroid (mi)
- C_t, C_p = coefficients

Clark Unit Hydrograph:

- Uses time–area curve and linear reservoir routing.
- R = storage coefficient.

5.5 Stream Gauging

Velocity methods:

- Two-point: $V = (V_{0.2} + V_{0.8})/2$
- Six-tenths: $V = V_{0.6}$
- Three-point: $V = (V_{0.2} + V_{0.6} + V_{0.8})/4$
- Five-point: $V = (V_{\text{surf}} + 3V_{0.2} + 3V_{0.6} + 2V_{0.8} + V_{\text{bed}})/10$
- Six-point: $V = (V_{\text{surf}} + 2V_{0.2} + 2V_{0.4} + 2V_{0.6} + 2V_{0.8} + V_{\text{bed}})/10$
- Two-tenths: $V = V_{0.2}$
- Surface velocity: $V = 0.87V_{\text{surf}}$

Discharge:

$$Q = \sum A_i V_i$$

5.6 Stormwater Management

Detention and Retention

- Detention: temporarily stores runoff, releases slowly.
- Retention: stores runoff permanently (no release).
- Infiltration systems: recharge groundwater.

- Swales: vegetated channels for conveyance and treatment.

Storage volume:

$$V_s = \max(V_{\text{in}} - V_{\text{out}})$$

Modified Puls Routing:

$$\frac{I_1 + I_2}{2}\Delta t + \left(\frac{S_1}{\Delta t} - \frac{O_1}{2}\right) = \frac{S_2}{\Delta t} + \frac{O_2}{2}$$

Erosion and Sediment Control

RUSLE:

$$A = RKLSCP$$

where:

- A = soil loss (tons/acre-yr)
- R = rainfall erosivity
- K = soil erodibility
- L = slope length
- S = slope steepness
- C = cover management
- P = conservation practice

Mulch factors and slope length limits: see handbook tables. Flag local requirements.

6. Construction Applications and Quality Control

- Temporary drainage and dewatering during construction.
- Erosion and sediment control plans and BMPs.
- Stormwater Pollution Prevention Plans (SWPPP).
- Inspection, maintenance, and specification conformance.

- Common field/design errors:
  - Inadequate inlet capacity
  - Poor grading
  - Sediment clogging
  - Undersized culverts
  - Missing energy dissipators
  - Failure to inspect HGL
  - Inadequate temporary storage

7. Common Errors and Exam Traps

Trap	Correction
Unit inconsistency	Use 1.486 in USCS Manning; 1.0 in SI
Confusing inlet/outlet control	Compute both; higher HW controls
Incorrect partial-full pipe use	Use hydraulic elements graph; max Q not at full
Neglecting tailwater	Always include TW in outlet control
Misusing Rational Method	i must correspond to t_c and return period
Forgetting minor losses	Add entrance, exit, bend, junction losses
Overlooking detention routing	Use Modified Puls; check storage
Misreading IDF curves	Match duration to t_c
Using wrong CN	Check land use and soil group
Forgetting K_s in inlet control	Apply slope correction
Using wrong D in Froude	$D = A/T$, not depth for non-rectangular

Trap	Correction
Ignoring composite roughness	Use weighted formula for multi-material culverts

8. Worked Examples

Example 1 — Manning's Equation: Normal Depth

Problem: Trapezoidal channel, bottom width $b = 6$ ft, side slopes $z = 2$ H:1 V, $n = 0.025$, $S = 0.001$, $Q = 120$ cfs. Find normal depth y_n.

Solution:

$$A = (b + zy)y = (6 + 2y)y$$

$$P = b + 2y\sqrt{1 + z^2} = 6 + 2y\sqrt{5} = 6 + 4.472y$$

$$R_H = A/P$$

$$Q = \frac{1.486}{n} A R_H^{2/3} S^{1/2}$$

Trial $y = 3.25$ ft:

$$A = (6 + 6.5)(3.25) = 40.625 \text{ ft}^2$$

$$P = 6 + 4.472(3.25) = 20.53 \text{ ft}$$

$$R_H = 40.625/20.53 = 1.979 \text{ ft}$$

$$R_H^{2/3} = 1.576$$

$$Q = \frac{1.486}{0.025}(40.625)(1.576)(0.001)^{0.5} = 59.44 \times 40.625 \times 1.576 \times 0.03162 \approx 119.9 \text{ cfs}$$

Close to 120 cfs.

Answer: $y_n \approx 3.25$ ft.

Example 2 — Critical Depth and Froude Number

Problem: Rectangular channel, $b = 10$ ft, $Q = 200$ cfs. Find y_c, $E_{\min}$, and classify flow at $y = 3$ ft.

Solution:

$$q = Q/b = 200/10 = 20 \text{ cfs/ft}$$

$$y_c = \left(\frac{q^2}{g}\right)^{1/3} = \left(\frac{20^2}{32.2}\right)^{1/3} = (12.42)^{1/3} = 2.32 \text{ ft}$$

$$E_{\min} = \frac{3}{2} y_c = 1.5(2.32) = 3.48 \text{ ft}$$

At $y = 3$ ft:

$$A = 10(3) = 30 \text{ ft}^2$$

$$V = 200/30 = 6.67 \text{ ft/s}$$

$$Fr = \frac{V}{\sqrt{gy}} = \frac{6.67}{\sqrt{32.2(3)}} = \frac{6.67}{9.83} = 0.68$$

$Fr < 1$, so flow is subcritical.

Answer: $y_c = 2.32$ ft, $E_{\min} = 3.48$ ft, subcritical at $y = 3$ ft.

Example 3 — Hydraulic Jump

Problem: Rectangular channel, upstream depth $y_1 = 1.0$ ft, upstream velocity $V_1 = 20$ ft/s. Find downstream depth y_2 and energy loss ΔE.

Solution:

$$Fr_1 = \frac{V_1}{\sqrt{g y_1}} = \frac{20}{\sqrt{32.2(1)}} = 3.52$$

$$\frac{y_2}{y_1} = \frac{1}{2}\left(\sqrt{1 + 8Fr_1^2} - 1\right) = \frac{1}{2}\left(\sqrt{1 + 8(12.4)} - 1\right) = \frac{1}{2}(10.01 - 1) = 4.50$$

$$y_2 = 4.50(1.0) = 4.50 \text{ ft}$$

$$\Delta E = \frac{(y_2 - y_1)^3}{4 y_1 y_2} = \frac{(4.50 - 1.0)^3}{4(1.0)(4.50)} = \frac{42.875}{18} = 2.38 \text{ ft}$$

Answer: $y_2 = 4.50$ ft, $\Delta E = 2.38$ ft.

Example 4 — Rational Method

Problem: Watershed area 25 acres. 15 acres $C = 0.30$, 10 acres $C = 0.70$. $t_c = 20$ min. 10-yr IDF gives $i = 4.2$ in./hr. Find peak flow.

Solution:

$$C_w = \frac{15(0.30) + 10(0.70)}{25} = \frac{4.5 + 7.0}{25} = 0.46$$

$$Q = C_w i A = 0.46(4.2)(25) = 48.3 \text{ cfs}$$

Answer: $Q \approx 48.3$ cfs.

Example 5 — Culvert Inlet Control

Problem: Circular concrete culvert, $D = 3$ ft, $Q = 120$ cfs, square edge with headwall. Use HDS-5 constants. Find inlet-control headwater.

Solution:

$$A = \frac{\pi D^2}{4} = \frac{\pi (3)^2}{4} = 7.069 \text{ ft}^2$$

$$\frac{Q}{AD^{0.5}} = \frac{120}{7.069\sqrt{3}} = \frac{120}{12.24} = 9.80$$

$$\left(\frac{Q}{AD^{0.5}}\right)^2 = 96.0$$

Unsubmerged:

$$\frac{HW}{D} = 0.0098(96.0) = 0.941$$

$$HW = 0.941(3) = 2.82 \text{ ft}$$

Submerged:

$$\frac{HW}{D} = 0.0398(96.0) + 0.67 = 3.82 + 0.67 = 4.49$$

$$HW = 4.49(3) = 13.47 \text{ ft}$$

Since unsubmerged gives $HW/D < 4$, use unsubmerged.

Answer: $HW_i \approx 2.82$ ft.

Example 6 — Culvert Outlet Control

Problem: Same culvert: $D = 3$ ft, $L = 200$ ft, $n = 0.012$, $k_e = 0.5$, $Q = 120$ cfs, $TW = 3$ ft, $S_0 = 0.001$. Find outlet-control headwater.

Solution:
Full flow:

$$V = Q/A = 120/7.069 = 16.98 \text{ ft/s}$$

$$R_H = D/4 = 0.75 \text{ ft}$$

$$H = \left(1 + k_e + \frac{29n^2L}{R_H^{4/3}}\right)\frac{V^2}{2g}$$

$$\frac{29n^2L}{R_H^{4/3}} = \frac{29(0.012)^2(200)}{0.75^{1.333}} = \frac{0.835}{0.682} = 1.225$$

$$1 + k_e + 1.225 = 1 + 0.5 + 1.225 = 2.725$$

$$\frac{V^2}{2g} = \frac{16.98^2}{64.4} = 4.48 \text{ ft}$$

$$H = 2.725(4.48) = 12.21 \text{ ft}$$

$$HW = H + TW - LS_0 = 12.21 + 3 - 200(0.001) = 12.21 + 3 - 0.2 = 15.01 \text{ ft}$$

Answer: $HW_o \approx 15.0$ ft. Outlet controls.

Example 7 — Gutter Flow

Problem: Triangular gutter, $S_x = 0.02$, $S_L = 0.01$, $n = 0.016$, spread $T = 8$ ft. Find Q, d, and V.

Solution:

$$Q = \frac{0.56}{n} S_x^{1.67} S_L^{0.5} T^{2.67}$$

$$S_x^{1.67} = 0.02^{1.67} = 0.00146$$

$$S_L^{0.5} = 0.1$$

$$T^{2.67} = 8^{2.67} = 257$$

$$Q = \frac{0.56}{0.016}(0.00146)(0.1)(257) = 1.31 \text{ cfs}$$

$$d = TS_x = 8(0.02) = 0.16 \text{ ft}$$

$$A = \frac{Td}{2} = \frac{8(0.16)}{2} = 0.64 \text{ ft}^2$$

$$V = Q/A = 1.31/0.64 = 2.05 \text{ ft/s}$$

Answer: $Q = 1.31$ cfs, $d = 0.16$ ft, $V = 2.05$ ft/s.

Example 8 — NRCS Runoff

Problem: $P = 4$ in., $CN = 80$. Find runoff Q.

Solution:

$$S = \frac{1000}{80} - 10 = 12.5 - 10 = 2.5 \text{ in.}$$

$$Q = \frac{(P - 0.2S)^2}{P + 0.8S} = \frac{(4 - 0.5)^2}{4 + 2.0} = \frac{12.25}{6.0} = 2.04 \text{ in.}$$

Answer: $Q = 2.04$ in.

9. High-Yield Checklist

□ Manning's equation: USCS vs. SI constant

□ $R_H = A/P$

□ $Fr = V/\sqrt{gD}$, $D = A/T$

□ Critical depth rectangular: $y_c = (q^2/g)^{1/3}$

□ Specific energy: $E = y + V^2/2g$

□ Hydraulic jump: $y_2/y_1 = 0.5(\sqrt{1 + 8Fr_1^2} - 1)$

□ GVF profiles: M, S, C, H, A

□ Circular pipe partial flow: max Q at $d/D \approx 0.94$

□ USGS culvert Types 1–6

□ Inlet control: unsubmerged vs. submerged

□ Outlet control: $H = H_e + H_f + H_o + H_b + H_j + H_g$

□ $HW = \max(HW_i, HW_o)$

□ Rational Method: $Q = CiA$

□ Composite C

- □ NRCS: $Q = (P - 0.2S)^2/(P + 0.8S)$, $S = 1000/CN - 10$
- □ t_c: sheet flow, SCS lag, Kirpich
- □ IDF: i for t_c and return period
- □ Curb/gutter: $Q = (0.56/n) S_x^{1.67} S_L^{0.5} T^{2.67}$
- □ Detention routing: Modified Puls
- □ RUSLE: $A = RKLSCP$
- □ HGL below gutter line
- □ Minimum/maximum velocities
- □ Units, units, units

10. Self-Test

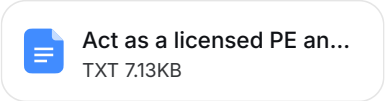
1. What is the USCS Manning constant?
2. For a rectangular channel, $q = 15$ cfs/ft. Find y_c.
3. $Fr = 2.0$, $y_1 = 1.5$ ft. Find y_2.
4. What is the maximum discharge ratio in a circular pipe?
5. What are the USGS culvert Types 1–6?
6. When does inlet control govern?
7. What is the Rational Method formula?
8. $P = 5$ in., $CN = 75$. Find Q.
9. What is the SCS lag formula?
10. What is the gutter flow equation for a triangular gutter?
11. What is the modified Puls routing equation?
12. What is RUSLE?
13. What is the minimum velocity in a storm sewer?
14. What is the difference between detention and retention?
15. What is the Froude number for a rectangular channel?
16. What is the specific energy equation?

17. What is the hydraulic radius?
18. What is the critical depth equation for a rectangular channel?
19. What is the energy loss in a hydraulic jump?
20. What is the composite Manning's n formula?

Answers:

1. 1.486
2. $y_c = (15^2/32.2)^{1/3} = 1.91$ ft
3. $y_2 = 0.5(\sqrt{1+8(4)}-1)(1.5) = 0.5(5.74)(1.5) = 4.31$ ft
4. $Q/Q_{\text{full}} \approx 1.08$ at $d/D \approx 0.94$
5. Type 1: steep, inlet control; Type 2: mild, outlet control; Type 3: tranquil, outlet control; Type 4: submerged outlet, outlet control; Type 5: rapid inlet, inlet control; Type 6: full free outfall, inlet control
6. When $HW_i > HW_o$
7. $Q = CiA$
8. $S = 1000/75 - 10 = 3.33$; $Q = (5-0.666)^2/(5+2.667) = 18.78/7.667 = 2.45$ in.
9. $t_L = 0.000526(1000/CN - 9)^{0.7} L^{0.8}/S^{0.5}$
10. $Q = (0.56/n) S_x^{1.67} S_L^{0.5} T^{2.67}$
11. $(I_1 + I_2)\Delta t/2 + (S_1/\Delta t - O_1/2) = S_2/\Delta t + O_2/2$
12. Revised Universal Soil Loss Equation: $A = RKLSCP$
13. 2 ft/s
14. Detention temporarily stores and releases; retention stores permanently
15. $Fr = V/\sqrt{gy}$
16. $E = y + V^2/2g$
17. $R_H = A/P$
18. $y_c = (q^2/g)^{1/3}$
19. $\Delta E = (y_2 - y_1)^3/(4y_1 y_2)$
20. $n_c = [\sum(p_i n_i^{1.5})/P]^{2/3}$

Final Note: This review is aligned with the NCEES PE Civil: Construction exam specifications and the *PE Civil Reference Handbook*, Version 2.0. If a detail is not specified by NCEES or falls outside the stated scope, it has been flagged. Use this material as exam-focused preparation designed to maximize readiness; it does not guarantee a passing score.



Thought for 2 seconds

> The user wants a comprehensive exam-focused review of "Storm Characteristics and Runoff Analysis" within the Hydraulics and Hydrology area of the NCEES PE Civil: Construction exam. They've provided a detailed prompt with specific subtopics to cover, aligned with the PE Civil Reference Handbook Version 2.0.
>
> Let me organize this into a thorough review document covering:
>
> 1. Storm Characteristics and Frequency
> 2. Runoff Analysis Methods
> 3. Types of Flow and Flow Characteristics
> 4. Time of Concentration
> 5. Hydrograph Development and Applications
> 6. Rainfall Measurement and Distribution
> 7. Erosion Control
> 8. Construction Applications and Quality Control
> 9. Common Errors and Exam Traps
> 10. High-Yield Checklist and Self-Test
>
> I need to reference the PE Civil Reference Handbook Version 2.0 content that was provided in the earlier file, particularly Section 6.5 Hydrology and related sections.
>
> Let me draft this comprehensively.

PE Civil: Construction — Storm Characteristics and Runoff Analysis

Exam-Focused Review for the NCEES PE Civil: Construction Exam

Based on NCEES PE Civil Construction exam specifications and the *PE Civil Reference Handbook*, Version 2.0.

Scope: Only Storm Characteristics and Runoff Analysis within Hydraulics and Hydrology. Related open-channel flow, hydraulics, erosion control, and construction topics are included only as they directly support storm characteristics and runoff analysis.

> **Disclaimer:** This review is exam-preparation material designed to maximize readiness. It does not guarantee a passing score. Always follow the current NCEES exam specifications and the supplied *PE Civil Reference Handbook* during the exam. If a constant, chart, or local criterion is not supplied by NCEES or in the exam question, do not invent it—flag it and use only what is provided.

0. How to Use This Review

Priority	Topic	Why It Matters
1	Rational Method + time of concentration	Most common peak-flow calculation
2	NRCS/SCS runoff method + curve numbers	Very common for watershed runoff depth/peak
3	IDF curves + storm frequency	Required input for almost all runoff problems
4	Unit hydrograph theory + SCS/Snyder	Hydrograph development questions
5	Rainfall averaging (Thiessen, isohyetal)	Gauging and distribution questions
6	RUSLE + erosion control	Construction-specific erosion questions
7	Common errors and unit traps	Prevents missed “easy” points

Exam strategy:

- Write units on every line.
- Check whether the question asks for peak flow, runoff depth, or hydrograph.
- For Rational Method, ensure i corresponds to t_c and the correct return period.
- For NRCS, ensure CN matches land use and soil group.

- For hydrographs, separate base flow from direct runoff.
- For erosion, ensure all RUSLE factors use consistent units.

1. Storm Characteristics and Frequency

1.1 Return Period, Recurrence Interval, and Annual Exceedance Probability

Average Recurrence Interval (ARI) or Return Period:

$$T = \frac{1}{p}$$

where:

- T = return period (years)
- p = probability of a single occurrence in a given storm period (annual exceedance probability)

Annual Exceedance Probability (AEP):

$$p = \frac{1}{T}$$

Example: A 25-year storm has $p = 1/25 = 0.04 = 4\%$ chance of being exceeded in any given year.

1.2 Risk and Reliability

Risk (probability of at least one exceedance in n years):

$$P(\text{at least once in } n \text{ years}) = 1 - (1 - p)^n$$

$$= 1 - \left(1 - \frac{1}{T}\right)^n$$

Reliability (probability of no exceedance in n years):

$$P(\text{nonexceedance each year for } n \text{ years}) = (1-p)^n$$

$$= \left(1 - \frac{1}{T}\right)^n$$

Example: For a 10-year storm over 5 years:

$$\text{Risk} = 1 - \left(1 - \frac{1}{10}\right)^5 = 1 - 0.9^5 = 1 - 0.590 = 0.410 = 41\%$$

$$\text{Reliability} = 0.9^5 = 0.590 = 59\%$$

> **Exam trap:** Do not confuse return period with "the storm that occurs every T years." It is a statistical average; two 100-year storms can occur in consecutive years.

1.3 Binomial Distribution for Flood Frequency

Probability of exactly k events in n trials:

$$f(k; n, p) = \binom{n}{k} p^k (1-p)^{n-k}$$

where:

$$\binom{n}{k} = \frac{n!}{k!(n-k)!}$$

- p = probability of an event in any one trial
- k = number of desired events
- n = number of trials

Example: Probability of exactly 2 ten-year storms in 10 years:

$$p = 0.10, \quad n = 10, \quad k = 2$$

$$f(2;10,0.10) = \binom{10}{2}(0.10)^2(0.90)^8 = 45 \times 0.01 \times 0.4305 = 0.194 = 19.4\%$$

1.4 Simplified Flood Frequency Equation

$$x_T = \bar{x} + K_T s$$

where:

- x_T = estimated event magnitude for return period T
- $\bar{x}$ = sample mean (observed data)
- K_T = frequency factor (Normal, Extreme Value, Exponential/Pearson)
- s = standard deviation of the sample

Frequency factor for Normal distribution:

Exceedance Probability	Return Period T	K_T
0.0001	10,000	3.719
0.001	1,000	3.090
0.005	200	2.576
0.010	100	2.326
0.025	40	1.960
0.050	20	1.645
0.100	10	1.282
0.200	5	0.842
0.500	2	0.000
0.900	1.11	–1.282

Exceedance Probability	Return Period T	K_T
0.950	1.053	–1.645
0.990	1.010	–2.326

> **Exam tip:** If the question provides a frequency factor table or chart, use it. If not, use the Normal distribution K_T values.

1.5 Rainfall Intensity–Duration–Frequency (IDF)

Point precipitation equations:

Equation 1:

$$i = \frac{cT^e}{T_d^m}$$

Equation 2:

$$i = \frac{cT^e}{(T_d + f)^m}$$

Equation 3:

$$i = \frac{cT^e}{T_d^m + f}$$

where:

- i = design rainfall intensity (in./hr)
- T_d = duration of event (min)
- T = return period (years)
- c, e, f, m = coefficients

Procedure for using IDF curves:

1. Determine time of concentration t_c.

2. Set duration $T_d = t_c$.
3. Read i from IDF curve for design return period.
4. Use i in Rational Method.

> **Exam trap:** The duration used in IDF must equal the time of concentration for the Rational Method. Do not use a different duration.

1.6 Design Storm Selection and Duration

- **Design storm:** The rainfall event used for design (e.g., 10-yr, 25-yr, 100-yr).
- **Duration:** Usually equal to t_c for peak flow; may be longer for storage design.
- **Storm types:** Convective, frontal, orographic, tropical.
- **Rainfall distribution patterns:** SCS Type I, IA, II, III; Huff distributions; local criteria.

> **Flag:** Specific design storm criteria (return period, duration, distribution) are often set by local drainage manuals. Use only what is supplied in the exam question.

1.7 Common Errors, Assumptions, and Exam Traps

Trap	Correction
Confusing ARI, AEP, return period	$T = 1/p$, $p = 1/T$
Forgetting n-year risk formula	Risk = $1 - (1 - p)^n$
Using wrong K_T	Match frequency distribution and return period
Misreading IDF curves	Match duration to t_c and return period
Using 100-yr storm as “safe”	It is a statistical average, not a guarantee

2. Runoff Analysis Methods

2.1 Rational Method

$$Q = CiA$$

where:

- Q = peak discharge (cfs)
- C = runoff coefficient
- i = rainfall intensity (in./hr) for duration t_c
- A = drainage area (acres)

Unit conversion: 1 acre-in./hr ≈ 1.008 cfs, usually approximated to 1.0.

Composite runoff coefficient:

$$C_w = \frac{\sum C_i A_i}{\sum A_i}$$

Runoff coefficients:

Land Use	C
Downtown business	0.70–0.95
Neighborhood business	0.50–0.70
Single-family residential	0.30–0.50
Multi-family detached	0.40–0.60
Multi-family attached	0.60–0.75
Suburban	0.25–0.40
Apartment	0.50–0.70
Light industrial	0.50–0.80
Heavy industrial	0.60–0.90

Land Use	C
Parks, cemeteries	0.10–0.25
Playgrounds	0.20–0.40
Railroad yards	0.20–0.40
Unimproved areas	0.10–0.30
Lawns, sandy soil, flat	0.05–0.10
Lawns, sandy soil, average	0.10–0.15
Lawns, sandy soil, steep	0.15–0.20
Lawns, heavy soil, flat	0.13–0.17
Lawns, heavy soil, average	0.18–0.22
Lawns, heavy soil, steep	0.25–0.35
Asphaltic streets	0.70–0.95
Concrete streets	0.80–0.95
Brick streets	0.70–0.85
Drives and walks	0.75–0.85
Roofs	0.75–0.95

Limitations:

- Valid for small watersheds, typically < 200 acres.
- Assumes uniform rainfall over the watershed.
- Peak flow only; no hydrograph.
- C and i must correspond to the same return period.

2.2 NRCS (SCS) Rainfall-Runoff Method

Runoff depth:

$$Q = \frac{(P - 0.2S)^2}{P + 0.8S}$$

$$S = \frac{1000}{CN} - 10$$

where:

- Q = runoff depth (in.)
- P = precipitation (in.)
- S = maximum basin retention (in.)
- CN = curve number

Initial abstraction:

$$I_a = 0.2S$$

Loss:

$$\text{Loss} = P - Q$$

When $P \leq I_a$: $Q = 0$.

SCS hydrologic soil groups:

Group	Description	Minimum Infiltration Rate (in./hr)
A	Deep sand, deep loess, aggregated silts	0.30–0.45
B	Shallow loess, sandy loam	0.15–0.30
C	Clay loams, shallow sandy loam, low organic content, high clay	0.05–0.15
D	Swelling clays, heavy plastic clays, saline soils	0.00–0.05

Typical CN values:

Land Use	A	B	C	D
Paved parking, roofs	98	98	98	98
Streets, paved w/ curbs	98	98	98	98
Gravel streets	76	85	89	91
Dirt streets	72	82	87	89
Commercial/business	89	92	94	95
Industrial	81	88	91	93
Residential 1/8 ac	77	85	90	92
Residential 1/4 ac	61	75	83	87
Residential 1/3 ac	57	72	81	86
Residential 1/2 ac	54	70	80	85
Residential 1 ac	51	68	79	84
Residential 2 ac	46	65	77	82
Newly graded area	77	86	91	94
Pasture, poor	68	79	86	89
Pasture, fair	49	69	79	84
Pasture, good	39	61	74	80
Woods, poor	45	66	77	83
Woods, fair	36	60	73	79
Woods, good	25	55	70	77

SCS Peak Discharge Method:

$$q_p = q_u A_m Q$$

where:

- q_p = peak flow (cfs)
- q_u = unit peak flow (cfs/mi²/in.)
- A_m = basin area (mi²)
- Q = runoff depth (in.)

$$\log q_u = C_0 + C_1 \log t_c + C_2 (\log t_c)^2$$

where C_0, C_1, C_2 = TR-55 coefficients.

2.3 Kinematic-Wave Runoff Method

$$Q = \alpha A^m$$

$$\alpha = \frac{1.486}{n} S_0^{1/2} P_c^{2/3}$$

where:

- $m = 5/3$ for wide rectangular/trapezoidal
- $m = 1.0$ for circular pipe
- $m = 1.33$ for triangular
- P_c = channel parameter (width, diameter, etc.)
- n = Manning's roughness

Time to equilibrium:

$$t_c = \frac{nL}{1.49 S_0^{1/2} h^{2/3}}$$

Recommended Manning's n for overland flow:

Cover	n
Concrete or asphalt	0.011
Bare sand	0.01
Graveled surface	0.02
Bare clay loam	0.02
Fallow, no residue	0.05
Chisel plow	0.07–0.40
Disk/harrow	0.08–0.30
No till	0.04–0.30
Range (natural)	0.13
Grass (bluegrass sod)	0.45
Dense grass	0.24
Bermuda grass	0.41

2.4 USGS Regression Method

Rural equations:

$$RQ_T = aA^bB^cC^d$$

where:

- RQ_T = T-year rural peak flow (cfs)
- a = regression constant
- b, c, d = regression coefficients

- A, B, C = basin characteristics

Urban equations: More complex; include basin development factor, impervious area, etc.

> **Flag:** USGS regression equations are region-specific. Use only if supplied in the exam question.

2.5 Comparison of Methods

Method	Best For	Limitations
Rational	Small watersheds, peak flow	No hydrograph, uniform rainfall
NRCS/SCS	Watershed runoff depth and peak	Requires CN, not for very large watersheds
Kinematic wave	Overland flow, time to equilibrium	Simplified hydraulics
USGS regression	Regional peak flow estimation	Region-specific equations

2.6 Common Errors, Assumptions, and Exam Traps

Trap	Correction
Misusing Rational Method	i must correspond to t_c and return period
Wrong CN	Check land use and soil group
Forgetting I_a	$Q = 0$ if $P \leq 0.2S$
Using S in inches vs. mm	Check units
Confusing Q (depth) with q_p (flow)	NRCS gives depth; peak method gives flow
Wrong m in kinematic wave	Match channel shape

3. Types of Flow and Flow Characteristics

3.1 Sheet Flow, Shallow Concentrated Flow, and Channel Flow

Sheet flow:

- Overland flow before concentration.
- Shallow depths, raindrop impact, high roughness.
- Use kinematic wave sheet flow formula.

Shallow concentrated flow:

- Flow in rills, gullies, or small channels.
- Use velocity versus slope graph or Manning's equation.

Channel flow:

- Established channels, streams, or pipes.
- Use Manning's equation and travel time.

3.2 Overland Flow Roughness Coefficients

See kinematic wave section for Manning's n for overland flow.

3.3 Velocity Versus Slope Relationships for Shallow Concentrated Flow

Use the NRCS velocity versus slope graph (Figure 15-4 in NEH Part 630). The graph gives velocity for:

- Forest with heavy ground litter
- Minimum tillage cultivation
- Short-grass pasture

- Cultivated straight row crops
- Nearly bare and untilled
- Grassed waterways
- Pavement and small upland gullies

> **Exam tip:** If the graph is supplied, read velocity directly. If not, use Manning's equation with appropriate n and R_H.

3.4 Open-Channel Flow Concepts as Applied to Runoff

Normal depth: depth for uniform flow; solve Manning implicitly.

Critical depth: depth at minimum specific energy; $Fr = 1$.

Froude number:

$$Fr = \frac{V}{\sqrt{gD}}$$

where:

$$D = \frac{A}{T}$$

- $Fr < 1$: subcritical
- $Fr = 1$: critical
- $Fr > 1$: supercritical

Hydraulic radius:

$$R_H = \frac{A}{P}$$

Channel geometry effects:

- Larger R_H → higher velocity and capacity.
- Wide shallow channels have lower R_H than narrow deep channels.

- Vegetation increases roughness, reduces velocity.

3.5 Common Errors, Assumptions, and Exam Traps

Trap	Correction
Confusing sheet flow with channel flow	Sheet flow uses kinematic wave; channel flow uses Manning
Using wrong n for overland flow	Match cover type
Forgetting $D = A/T$ in Froude	Not just depth for non-rectangular
Using normal depth for critical flow	They are different

4. Time of Concentration

4.1 Definition and Importance

Time of concentration t_c: Time for runoff to travel from the hydraulically most distant point of the watershed to the point of interest.

Importance:

- Determines duration for IDF intensity in Rational Method.
- Determines time to peak in unit hydrographs.
- Affects peak flow magnitude.

4.2 SCS Lag Formula

$$t_L = \frac{0.000526(1000/CN - 9)^{0.7} L^{0.8}}{S^{0.5}}$$

$$t_c = \frac{5}{3} t_L$$

where:

- t_L = lag time (hr)
- L = watershed length (ft)
- S = watershed slope (%)
- CN = curve number

4.3 Sheet Flow Formula (Kinematic Wave)

$$T_t = \frac{K_u (nL)^{0.8}}{(P_2)^{0.5} S^{0.4}}$$

where:

- T_t = sheet flow travel time (min)
- $K_u = 0.933$ (USCS)
- n = Manning's roughness for overland flow
- L = flow length (ft)
- P_2 = 2-yr, 24-hr rainfall (in.)
- S = slope (ft/ft)

Limitations: Sheet flow equation is valid for flow lengths up to 100 ft (sometimes 300 ft). Use shallow concentrated flow for longer lengths.

4.4 Shallow Concentrated Flow Travel Time

$$T_t = \frac{L}{V}$$

where V is from the velocity versus slope graph or Manning's equation.

4.5 Channel Flow Travel Time

$$T_t = \frac{L}{V}$$

where V is from Manning's equation.

4.6 Inlet Flow Formula and Overland Flow Time

$$t_i = C\left(\frac{L}{Si^2}\right)^{1/3}$$

where:

- t_i = time of overland flow (min)
- L = distance of overland flow (ft)
- S = slope of land (ft/ft)
- i = rainfall intensity (in./hr)
- C = coefficient

Total time of concentration:

$$t = t_i + t_s$$

where t_s = time of sewer flow.

4.7 Watershed Length, Slope, and Curve Number Effects

- Longer L → larger t_c
- Steeper S → smaller t_c
- Higher CN → smaller t_c (more impervious, faster runoff)

4.8 Common Errors, Assumptions, and Exam Traps

Trap	Correction
Using sheet flow for long distances	Sheet flow limited to ~100 ft
Forgetting to convert t_c to hours	SCS lag gives t_c in hours; Rational Method uses minutes
Using wrong P_2	Use 2-yr, 24-hr rainfall
Adding travel times incorrectly	Sum all segments: sheet + shallow + channel
Using t_c for wrong return period	t_c is independent of return period

5. Hydrograph Development and Applications

5.1 Unit Hydrograph Theory and Assumptions

Unit hydrograph: Direct runoff hydrograph resulting from one unit of runoff (1 in. or 1 mm) occurring uniformly in space and time over a specified duration.

Assumptions:

- Linear response (proportionality)
- Time invariance
- Uniform rainfall distribution
- Same time base for same duration

Direct runoff hydrograph: Total hydrograph minus base flow.

Base flow separation:

- Straight-line method
- Fixed-base method
- Variable-slope method

Effective rainfall: Rainfall excess that becomes direct runoff.

5.2 Snyder Synthetic Unit Hydrograph

$$t_p = C_t(LL_c)^{0.3}$$

$$q_p = \frac{640 C_p A}{t_p}$$

$$W_{50} = 2.5 q_p^{-1.08}$$

$$W_{75} = 1.22 q_p^{-1.08}$$

where:

- t_p = time to peak (hr)
- L = stream length from outlet to upstream limits (mi)
- L_c = stream length from outlet to centroid (mi)
- C_t, C_p = coefficients
- q_p = peak flow (cfs)
- A = drainage area (mi²)
- W_{50}, W_{75} = widths at 50% and 75% of peak (hr)

Standard duration:

$$t_D = \frac{t_p}{5.5}$$

Lag time:

$$t_{pR} = t_p + 0.25(t_r - t_D)$$

5.3 SCS (NRCS) Dimensionless Unit Hydrograph

$$q_p = \frac{484AQ}{t_p}$$

$$t_p = \frac{D}{2} + t_L$$

$$T_b = 2.67t_p$$

where:

- q_p = peak flow (cfs)
- A = area (mi²)
- Q = runoff depth (in.)
- t_p = time to peak (hr)
- D = duration of excess rainfall (hr)
- t_L = lag time (hr)
- T_b = time base (hr)

SCS lag method:

$$t_L = \frac{1}{1.7}t_c$$

or

$$t_c = \frac{5}{3}t_L$$

Dimensionless unit hydrograph ratios:

t/T_p	q/q_p	Q_a/Q
0	0.000	0.000
0.1	0.030	0.001
0.2	0.100	0.006
0.3	0.190	0.012

t/T_p	q/q_p	Q_a/Q
0.4	0.310	0.035
0.5	0.470	0.065
0.6	0.660	0.107
0.7	0.820	0.163
0.8	0.930	0.228
0.9	0.990	0.300
1.0	1.000	0.375
1.1	0.990	0.450
1.2	0.930	0.522
1.3	0.860	0.589
1.4	0.780	0.650
1.5	0.680	0.700
1.6	0.560	0.751
1.7	0.460	0.790
1.8	0.390	0.822
1.9	0.330	0.849
2.0	0.280	0.871
2.2	0.207	0.908
2.4	0.147	0.934
2.6	0.107	0.953

t/T_p	q/q_p	Q_a/Q
2.8	0.077	0.967
3.0	0.055	0.977
3.2	0.040	0.984
3.4	0.029	0.989
3.6	0.021	0.993
3.8	0.015	0.995
4.0	0.011	0.997
4.5	0.005	0.999
5.0	0.000	1.000

5.4 Clark Unit Hydrograph

Time–area method:

$$\frac{A_t}{A_c} = 1.414 \left(\frac{t}{t_c} \right)^{1.5} \quad \text{for } 0 \le \frac{t}{t_c} \le 0.5$$

$$\frac{A_t}{A_c} = 1 - 1.414 \left(1 - \frac{t}{t_c} \right)^{1.5} \quad \text{for } 0.5 \le \frac{t}{t_c} \le 1.0$$

Linear reservoir routing:

$$O_t = C I_t + (1 - C) O_{t-1}$$

$$C = \frac{\Delta t}{R + \Delta t}$$

where:

- A_t = contributing area at time t
- A_c = total watershed area
- t_c = time of concentration
- O_t = outflow at time t
- I_t = inflow at time t
- R = storage coefficient
- Δt = time step

5.5 Hydrograph Estimate of Stream Flow

$$Q_t = Q_0 K^t$$

where:

- Q_t = discharge at time t
- Q_0 = initial discharge at start of recession
- K = recession constant

5.6 Common Errors, Assumptions, and Exam Traps

Trap	Correction
Forgetting base flow separation	Separate direct runoff from total flow
Using wrong t_p formula	Snyder vs. SCS differ
Confusing t_p and t_L	t_p = time to peak; t_L = lag time
Using Q in inches vs. cfs	SCS peak equation uses Q in inches
Forgetting unit conversion	640 for Snyder, 484 for SCS
Misreading dimensionless hydrograph	Match t/T_p to q/q_p

6. Rainfall Measurement and Distribution

6.1 Precipitation Gauge Types and Analysis

Gauge types:

- Standard non-recording gauge
- Recording gauge (tipping bucket, weighing)
- Radar estimates
- Satellite estimates

Precipitation gauge analysis:

$$P = \frac{1}{n} \sum_{i=1}^{n} P_i$$

6.2 Arithmetic Mean Method

$$P = \frac{1}{n} \sum_{i=1}^{n} P_i$$

where:

- P = average precipitation
- P_i = precipitation at gauge i
- n = number of gauges

Best for: Flat, uniform terrain with evenly distributed gauges.

6.3 Thiessen Polygon Method

Procedure:

1. Plot stations on scaled map.
2. Connect adjacent stations with dashed lines.
3. Construct perpendicular bisectors.
4. Form polygons around each station.
5. Compute area of each polygon.
6. Multiply each area by its station rainfall.
7. Sum products and divide by total area.

$$P = \frac{\sum A_i P_i}{\sum A_i}$$

Best for: Uneven gauge distribution, variable terrain.

6.4 Isohyetal Method

Procedure:

1. Plot stations and rainfall values.
2. Draw contours of equal precipitation (isohyets).
3. Compute areas between successive isohyets.
4. Multiply each area by average of two isohyet values.
5. Sum and divide by total area.

$$P = \frac{\sum A_i P_i}{\sum A_i}$$

Best for: Irregular terrain, large watersheds.

6.5 Distance Weighting and Normal-Ratio Method

Distance weighting:

$$P = \frac{\sum_{i=1}^{n} \frac{P_i}{L_i^2}}{\sum_{i=1}^{n} \frac{1}{L_i^2}}$$

where L_i = distance from index station to station X_i.

Normal-ratio method:

$$P_x = \frac{1}{n} \sum_{i=1}^{n} \frac{A_x}{A_i} P_i$$

where:

- A_x = average annual catch at station x
- A_i = average annual catch at gauge i
- P_i = rainfall depth at gauge i

6.6 Regression Method (Double-Mass Curve)

Procedure:

1. Plot cumulative catch of questionable gauge vs. cumulative catch of surrounding gauges.
2. Check for slope changes.
3. Adjust using ratio of slopes:

$$y_1 = Y_i \frac{S_1}{S_2}$$

6.7 Estimation and Analysis for Missing Data

Arithmetic/Station Average Method:

$$P_x = \frac{1}{n} \sum_{i=1}^{n} P_i$$

Normal-Ratio Method:

$$P_x = \frac{1}{n} \sum_{i=1}^{n} \frac{A_x}{A_i} P_i$$

6.8 Rainfall Distribution Patterns and Design Storms

- **SCS Type I, IA, II, III:** Different temporal distributions.
- **Huff distributions:** Four quartiles.
- **Local criteria:** Use if supplied.

> **Flag:** Specific design storm distributions are often set by local drainage manuals. Use only what is supplied.

6.9 Common Errors, Assumptions, and Exam Traps

Trap	Correction
Using arithmetic mean for uneven gauges	Use Thiessen or isohyetal
Forgetting to scale map distances	Use consistent units
Misreading isohyets	Average between contours
Using wrong missing-data method	Normal-ratio if annual catches differ
Forgetting double-mass curve slope change	Check for inconsistency

7. Erosion Control

7.1 Revised Universal Soil Loss Equation (RUSLE)

$$A = RKLSCP$$

where:

- A = predicted average soil loss (tons/acre-yr)
- R = climatic erosivity
- K = soil erodibility
- L = slope length factor
- S = slope steepness factor
- C = crop and cover management factor
- P = conservation practice factor

K values (tons/acre):

Textural Class	Organic Matter 0.5%	Organic Matter 2%
Fine sand	0.16	0.14
Very fine sand	0.42	0.36
Loamy sand	0.12	0.10
Loamy very fine sand	0.44	0.38
Sandy loam	0.27	0.24
Very fine sandy loam	0.47	0.41
Silt loam	0.48	0.42
Clay loam	0.28	0.25
Silty clay loam	0.37	0.32
Silty clay	0.25	0.23

LS factor:

$$LS = \left(\frac{\lambda}{72.6}\right)^{m} (65.41 \sin^2\theta + 4.56 \sin\theta + 0.065)$$

where:

- λ = slope length (ft)
- m = 0.2 for <1%, 0.3 for 1–3%, 0.4 for 3.5–4.5%, 0.5 for ≥5%
- θ = angle of slope

Mulch factors and length limits:

Mulch Type	Rate (tons/acre)	Slope	C	Length Limit (ft)
None	0	all	1.0	–
Straw/hay, anchored	1.0	1–5%	0.20	200
Straw/hay, anchored	1.0	6–10%	0.20	100
Straw/hay, anchored	1.5	1–5%	0.12	300
Straw/hay, anchored	1.5	6–10%	0.12	150
Straw/hay, anchored	2.0	1–5%	0.06	400
Straw/hay, anchored	2.0	6–10%	0.06	200
Straw/hay, anchored	2.0	11–15%	0.07	150
Straw/hay, anchored	2.0	16–20%	0.11	100
Straw/hay, anchored	2.0	21–25%	0.14	75
Straw/hay, anchored	2.0	26–33%	0.17	50
Straw/hay, anchored	2.0	34–50%	0.20	35
Crushed stone	135	<16%	0.05	200
Crushed stone	135	16–20%	0.05	150

Mulch Type	Rate (tons/acre)	Slope	C	Length Limit (ft)
Crushed stone	135	21–33%	0.05	100
Crushed stone	135	34–50%	0.05	75
Wood chips	7	<16%	0.08	75
Wood chips	7	16–20%	0.08	50
Wood chips	12	<16%	0.05	150
Wood chips	12	16–20%	0.05	100
Wood chips	12	21–33%	0.05	75
Wood chips	25	<16%	0.02	200
Wood chips	25	16–20%	0.02	150
Wood chips	25	21–33%	0.02	100
Wood chips	25	34–50%	0.02	75

7.2 Sediment Control Principles and BMPs

- **Silt fences:** Intercept sheet flow; not for concentrated flow.
- **Sediment basins:** Capture sediment-laden runoff.
- **Check dams:** Reduce velocity in channels.
- **Mulching:** Protect soil surface.
- **Seeding:** Establish vegetation.
- **Erosion control blankets:** Protect slopes.
- **Storm drain inlet protection:** Prevent sediment entry.
- **SWPPP:** Stormwater Pollution Prevention Plan.

7.3 Common Errors, Assumptions, and Exam Traps

Trap	Correction
Forgetting one RUSLE factor	$A = RKLSCP$
Using wrong K	Match soil texture and organic matter
Using wrong C	Match cover type and slope
Forgetting length limits	Exceeding limit requires higher application rate
Using silt fence for concentrated flow	Silt fence is for sheet flow only

8. Construction Applications and Quality Control

8.1 Temporary and Permanent Stormwater Management

- **Temporary:** During construction; sediment control, diversions, dewatering.
- **Permanent:** Post-construction; detention/retention, infiltration, swales.

8.2 Erosion and Sediment Control Plans and SWPPP

- **ESC plan:** Erosion and sediment control measures.
- **SWPPP:** Stormwater Pollution Prevention Plan; required for construction sites.
- **BMPs:** Best Management Practices.

8.3 Detention and Retention Ponds, Infiltration Systems, and Swales

- **Detention:** Temporarily stores runoff, releases slowly.
- **Retention:** Stores runoff permanently.
- **Infiltration:** Recharges groundwater.
- **Swales:** Vegetated channels for conveyance and treatment.

8.4 Inspection, Maintenance, and Specification Conformance

- Regular inspection of BMPs.
- Maintenance of sediment basins, silt fences, etc.
- Conformance with specifications and permits.

8.5 Common Field and Design Errors

- Inadequate inlet capacity
- Poor grading
- Sediment clogging
- Undersized culverts
- Missing energy dissipators
- Failure to inspect HGL
- Inadequate temporary storage

9. Common Errors and Exam Traps (Summary)

Trap	Correction
Unit inconsistencies	Use consistent USCS or SI
Misapplication of Rational Method	i for t_c and return period
Incorrect CN selection	Match land use and soil group
Forgetting I_a	$Q = 0$ if $P \leq 0.2S$
Confusing ARI, AEP, return period	$T = 1/p$
Misreading IDF curves	Match duration to t_c
Neglecting base flow separation	Separate direct runoff

Trap	Correction
Overlooking erosion control	Use RUSLE and BMPs
Using sheet flow for long distances	Limit to ~100 ft
Forgetting unit conversion in SCS	Q in inches, q_p in cfs
Using wrong frequency factor	Match distribution
Adding travel times incorrectly	Sum all segments
Using wrong m in kinematic wave	Match channel shape
Forgetting $D = A/T$ in Froude	Not just depth
Using normal depth for critical flow	They are different

10. High-Yield Checklist

□ $T = 1/p$, $p = 1/T$

□ Risk = $1 - (1 - p)^n$

□ Reliability = $(1 - p)^n$

□ Binomial: $f(k; n, p) = \binom{n}{k} p^k (1 - p)^{n-k}$

□ $x_T = \bar{x} + K_T s$

□ IDF: $i = cT^e / T_d^m$

□ Rational Method: $Q = CiA$

□ Composite C

□ NRCS: $Q = (P - 0.2S)^2 / (P + 0.8S)$, $S = 1000/CN - 10$

□ SCS peak: $q_p = q_u A_m Q$

□ Kinematic wave: $Q = \alpha A^m$

□ t_c: sheet flow, SCS lag, Kirpich

□ Sheet flow: $T_t = K_u (nL)^{0.8} / (P_2^{0.5} S^{0.4})$

- [ ] SCS lag: $t_L = 0.000526(1000/CN - 9)^{0.7} L^{0.8} / S^{0.5}$
- [ ] $t_c = 5/3 t_L$
- [ ] Unit hydrograph: 1 in. runoff
- [ ] Snyder: $t_p = C_t (L L_c)^{0.3}$, $q_p = 640 C_p A / t_p$
- [ ] SCS: $q_p = 484 A Q / t_p$, $t_p = D/2 + t_L$
- [ ] Clark: time–area + linear reservoir
- [ ] Recession: $Q_t = Q_0 K^t$
- [ ] Thiessen: $P = \sum A_i P_i / \sum A_i$
- [ ] Isohyetal: $P = \sum A_i P_i / \sum A_i$
- [ ] Normal-ratio: $P_x = (1/n) \sum (A_x / A_i) P_i$
- [ ] RUSLE: $A = RKLSCP$
- [ ] $LS = (\lambda / 72.6)^m (65.41 \sin^2\theta + 4.56 \sin\theta + 0.065)$
- [ ] Mulch factors and length limits
- [ ] BMPs and SWPPP
- [ ] Units, units, units

11. Self-Test

1. What is the return period for a storm with 5% annual exceedance probability?
2. What is the risk of a 25-year storm over 10 years?
3. What is the probability of exactly 1 ten-year storm in 5 years?
4. What is the Rational Method formula?
5. $P = 6$ in., $CN = 70$. Find Q.
6. What is the SCS lag formula?
7. What is the sheet flow formula?
8. What is the Snyder q_p formula?
9. What is the SCS q_p formula?
10. What is the RUSLE equation?

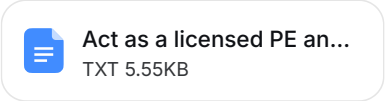
11. What is the Thiessen polygon formula?
12. What is the normal-ratio method?
13. What is the difference between detention and retention?
14. What is the difference between ARI and AEP?
15. What is the binomial distribution formula?
16. What is the frequency factor for a 50-year storm using Normal distribution?
17. What is the kinematic wave equation?
18. What is the time to equilibrium in kinematic wave?
19. What is the recession constant equation?
20. What is the LS factor in RUSLE?

Answers:

1. $T = 1/0.05 = 20$ years
2. Risk = $1 - (1 - 1/25)^{10} = 1 - 0.96^{10} = 1 - 0.665 = 0.335 = 33.5\%$
3. $f(1; 5, 0.10) = \binom{5}{1}(0.10)^1(0.90)^4 = 5 \times 0.10 \times 0.6561 = 0.328 = 32.8\%$
4. $Q = CiA$
5. $S = 1000/70 - 10 = 4.286$; $Q = (6 - 0.857)^2/(6 + 3.429) = 26.45/9.429 = 2.81$ in.
6. $t_L = 0.000526(1000/CN - 9)^{0.7} L^{0.8}/S^{0.5}$
7. $T_t = K_u(nL)^{0.8}/(P_2^{0.5} S^{0.4})$
8. $q_p = 640 C_p A/t_p$
9. $q_p = 484 AQ/t_p$
10. $A = RKLSCP$
11. $P = \sum A_i P_i / \sum A_i$
12. $P_x = (1/n)\sum (A_x/A_i) P_i$
13. Detention temporarily stores and releases; retention stores permanently
14. ARI = return period T; AEP = annual exceedance probability $p = 1/T$
15. $f(k; n, p) = \binom{n}{k} p^k (1-p)^{n-k}$
16. $K_T = 2.054$ (approximately, for $T = 50$, $p = 0.02$)
17. $Q = \alpha A^m$

18. $t_c = nL/(1.49S_0^{1/2}h^{2/3})$
19. $Q_t = Q_0K^t$
20. $LS = (\lambda/72.6)^m(65.41\sin^2\theta + 4.56\sin\theta + 0.065)$

Final Note: This review is aligned with the NCEES PE Civil: Construction exam specifications and the *PE Civil Reference Handbook*, Version 2.0. If a detail is not specified by NCEES or falls outside the stated scope, it has been flagged. Use this material as exam-focused preparation designed to maximize readiness; it does not guarantee a passing score.



Thought for 24 seconds

> We need answer. Need craft comprehensive exam-focused review of Detention/Retention Ponds only, aligned with NCEES PE Civil Construction, PE Civil Reference Handbook v2.0. User asks as licensed PE instructor. Need cover all subtopics, definitions, formulas, step-by-step procedures, worked examples, tips, tables, practice problems with detailed solutions, high-yield checklist, self-test. Must flag unspecified details. No guarantee. Need likely produce final answer. We need ensure accurate formulas from handbook. We can cite sections. We have handbook excerpt in previous context. Need use only provided. We can produce detailed but not overly huge? Desired oververbosity 5. But user asks comprehensive. We can make a robust review.
>
> Need only Detention/Retention Ponds. Include necessary hydrology. Use Section 6.5.9 etc. We need be careful with formulas. Let's recall from handbook excerpt:
>
> 6.5.9.1 Detention and Retention:
> Rational Method
> V = i A C t
> V_in = Q_in t
> V_out = Q_out t? Actually excerpt:
> V
> i ACt
> V
> Q t
> 0
> in
> out
> Σ

=
=
where
Q0 = outflow rate (cfs)
A = drainage area (acres)
C = rational method runoff coefficient
t = time (min)
i = average rainfall intensity (in./hr)
Storage volume required:
V
V
V
maximum
s
in
out
=
–
Routing Equation
I
I
t
S
O
t
S
O
t
2
1
2
1
2
1
1
1
1
2
2
2

Δ
Δ
Δ
+
+
−
=
+
–
c
c
i
m
m
where
I = inflow (volume/time)
O = outflow (volume/time)
S = storage (volume/time)
Δt = time interval (time)
Modified Puls Routing Method
I
I
t
S
O
t
S
O
2
2
1
2
1
1
2
2
Δ
Δ
+
+

–
=
+
–
d
i
n

Actually formula in handbook: (I1+I2)/2 Δt + (S1/Δt - O1/2) = S2/Δt + O2/2. Good.

6.5.8 Depletions etc. 6.5.9.2 Erosion. RUSLE. Mulch factors. Not all needed but water quality/erosion.

Outlet structures: Weir flow Q=CLH^{3/2}; orifice Q=C_d A sqrt(2gh) or Q=C A sqrt(2gh). Need include. From handbook 6.2.6 Orifices: Q = C_d A0 sqrt(2gh)? We can present. Weirs: rectangular sharp crested Q = C L H^{3/2}; broad crested etc. Culvert outlet control: inlet/outlet control maybe refer.

Need include TR-55 storage volume estimation? The handbook didn't have explicit TR-55 storage equation? It has NRCS peak method. We can state TR-55 graphical method: storage volume estimated from ratio of storage volume to runoff volume vs peak outflow to peak inflow. But NCEES handbook may not include. Flag if not supplied. We can describe generally but say use if supplied.

Need include stage-storage, elevation-area-volume curves. We can give formulas: V = ∫ A dh, trapezoidal/prismoidal. Use average end area. Stage-discharge curves for outlet structures.

Need include design storms, IDF, time of concentration, inflow hydrograph development, unit hydrograph, CN, Rational. Critical storm duration.

Water quality volume: first flush, WQv. Sediment forebay. Maintenance.

Construction: embankment, liners, seepage, safety.

Common errors.

Practice problems. We can create several examples:

Example 1: Storage volume rational method. Given area 20 acres, C=0.4, IDF intensity for tc=30 min 10-yr = 3 in/hr, allowable outflow 5 cfs. Compute inflow peak, storage volume? Need use V=(Qin-Qout)*t? Actually storage volume approximate: For critical duration, V_s = (Q_in - Q_out) * t? Need use average? Better: For rational method, storage volume = (Qin - Qout)*t_c maybe. But if Q_in varies. Use modified rational: V = (Q_in - Q_out) * t_d where t_d is critical duration. If Q_in = C i A, Q_out = allowable. But i depends on duration. Need iterate. We can do simple problem using given average i and duration.

Example 2: Modified Puls routing step. Given inflow hydrograph, storage-outflow relationship, time step, initial conditions, compute outflow and storage at next step.

Example 3: Orifice/weir outlet. Given pond stage, orifice diameter, weir length, compute outflow at given head. Or design riser.

Example 4: NRCS runoff depth and pond storage. Given P, CN, area, compute runoff volume; then storage volume as fraction? Maybe use TR-55.

Example 5: RUSLE erosion for construction slope.

Need detailed solutions.

Need high-yield checklist and self-test.

Need align with NCEES. Flag if not specified.

Let's structure:

1. Scope and how to use.
2. Purpose and Types
3. Hydrology Inputs: Design storms, IDF, tc, runoff methods
4. Storage Volume Estimation
5. Routing Methods (Modified Puls)
6. Outlet Structures and Hydraulics
7. Water Quality and Environmental
8. Construction, Maintenance, Safety
9. Common Errors and Exam Traps
10. Worked Examples
11. High-Yield Checklist
12. Self-Test with answers

Need include tables: differences detention/retention, storage components, weir/orifice coefficients, routing table template, critical storm duration.

Need use formulas from handbook. Ensure units.

Let's craft. Need not overdo but comprehensive.

Potential issue: The user included previous file content? No, this is new conversation. We have context from previous? The assistant previously responded to similar tasks. We need answer now.

Let's write with Markdown, tables, bullets. Keep exam-focused.

Need maybe mention "If detail not specified by NCEES or outside scope, flag." So include flag notes.

Let's draft.

We need be accurate. Let's recall PE Civil Reference Handbook formulas:

- Rational Method Q = C i A.
- NRCS Q = (P - 0.2S)^2/(P+0.8S), S=1000/CN -10.

- Time of concentration: SCS lag, sheet flow, etc.
- Unit hydrograph: SCS qp=484 A Q / tp; Snyder.
- Routing: Modified Puls: (I1+I2)/2 Δt + (S1/Δt - O1/2) = S2/Δt + O2/2.
- Orifice: Q = C_d A sqrt(2gh). Weir: Q = C L H^{3/2}. Broad crested: Q = C L H^{3/2}? Actually broad crested Q = C L H^{3/2} with C around 3.0? In handbook 6.2.5.4: Q = C L H^{3/2}; C=3.33 for rectangular weirs; Rehbock. Sharp-crested with end contractions: Q = C [L - n(0.1H)] H^{3/2}. V-notch: Q = C H^{5/2} etc.
- Orifice: Q = C_d A0 sqrt(2gh) for free discharge.
- Culvert: inlet/outlet control.

Need include stage-storage: V = ∫ A dh; for trapezoidal pond, V = (A1 + A2)/2 * Δh or prismoidal.

Critical storm duration: for detention, peak outflow occurs for a range of durations; need check multiple durations to find max storage. Use Modified Rational Method: Q_in(t) = C i(t) A; Q_out assumed constant? Actually modified rational: storage volume V = (Q_in - Q_out) * t_d. Critical duration when dV/dt = 0? We can explain: For each duration, compute required storage; largest required storage governs.

TR-55 storage volume: graphical method: Vs/Vr = f(qo/qi). If not supplied, flag.

Water quality volume: WQv = 0.05 to 0.5 in per impervious acre? Typically first 0.5 inch of runoff. Use local criteria.

Let's write final. Ensure no guaranteed pass. End with disclaimer.

We can include practice problems with solutions. Let's do 5 examples.

Example 1: Rational storage. Area 15 ac, C=0.35, 25-yr IDF: i = 4.0 in/hr for 20 min, i=3.2 for 30 min, i=2.5 for 45 min. Allowable outflow 6 cfs. Determine critical duration and storage. Use Modified Rational: Q_in = C i A. For each duration, compute Q_in, then V=(Q_in-Q_out)*t (convert t to seconds? Q cfs, t sec → ft3). Need convert t min to sec. Compute:
*20 min: Q=0.35*4.0*15=21 cfs; V=(21-6)*20*60=18,000 ft3.
30 min: Q=0.35*3.2*15=16.8 cfs; V=(16.8-6)*30*60=19,440 ft3.
45 min: Q=0.35*2.5*15=13.125; V=(13.125-6)*45*60=19,237.5 ft3.
Critical 30 min, storage ~19,440 ft3. Good.

Example 2: Modified Puls routing. Given: Δt=10 min, at time 1: S1=20,000 ft3, O1=5 cfs, I1=10 cfs, I2=15 cfs. Compute S2/Δt + O2/2. Use equation: (I1+I2)/2 Δt + (S1/Δt - O1/2) = S2/Δt + O2/2.
Convert Δt=600 sec. (10+15)/2 *600 = 12.5*600=7,500 ft3. S1/Δt = 20,000/600=33.333 cfs. O1/2=2.5 cfs. LHS = 7500 + (33.333-2.5)*600? Wait careful: formula terms: (I1+I2)/2 Δt + (S1/Δt - O1/2) = S2/Δt + O2/2. But units: (I1+I2)/2 is cfs, times Δt sec = ft3. S1/Δt is cfs, O1/2 cfs. So LHS = 7500 + (33.333 - 2.5)*? No, S1/Δt - O1/2 is cfs, but formula as written: (I1+I2)/2 Δt + (S1/Δt - O1/2) = ... The second term should be multiplied by Δt? Let's check handbook:
Modified Puls Routing Method
I

I
t
S
O
t
S
O
2
2
1
2
1
1
2
2
Δ
Δ
+
+
−
=
+
−
d
i
n
Actually in handbook: \frac{I_1+I_2}{2}\Delta t + \left(\frac{S_1}{\Delta t} - \frac{O_1}{2}\right) = \frac{S_2}{\Delta t} + \frac{O_2}{2}
Wait units: S/Δt is cfs. O/2 cfs. So LHS terms: (I1+I2)/2 Δt is ft3, plus cfs? That can't be. The correct equation is:
(I1+I2)/2 Δt + (S1/Δt - O1/2) Δt = S2/Δt Δt + O2/2 Δt? Actually standard: (I1+I2)/2 Δt + (S1/Δt - O1/2) Δt = (S2/Δt + O2/2) Δt? No.
Let's derive: Storage equation: (I1+I2)/2 - (O1+O2)/2 = (S2-S1)/Δt.
Multiply by Δt: (I1+I2)/2 Δt - (O1+O2)/2 Δt = S2 - S1.
Rearrange: (I1+I2)/2 Δt + S1 - O1/2 Δt = S2 + O2/2 Δt.
Divide by Δt: (I1+I2)/2 + S1/Δt - O1/2 = S2/Δt + O2/2.
Ah handbook omitted Δt on second term? Actually if divide by Δt, the equation is:
(I1+I2)/2 + S1/Δt - O1/2 = S2/Δt + O2/2.
But handbook shows (I1+I2)/2 Δt + (S1/Δt - O1/2) = S2/Δt + O2/2. That seems inconsistent. Maybe they

multiplied by Δt but only first term? Let's not rely. We can state correct form: (I1+I2)/2 Δt + S1 - O1 Δt/2 = S2 + O2 Δt/2. Or use standard: (I1+I2)/2 Δt + (S1 - O1 Δt/2) = S2 + O2 Δt/2. Then compute.

Better to present standard routing equation: (I1+I2)/2 Δt - (O1+O2)/2 Δt = S2 - S1. Rearranged: (I1+I2)/2 Δt + S1 - O1 Δt/2 = S2 + O2 Δt/2. Then define storage indication. That's clear.

In example, compute S2 + O2 Δt/2 = LHS. Then use stage-storage-outflow to find O2 and S2.

Example: Δt=10 min=600s. I1=10, I2=15 cfs. S1=20,000 ft3. O1=5 cfs.
LHS = (10+15)/2 *600 + 20,000 - 5*600/2 = 12.5*600 + 20,000 - 1,500 = 7,500 + 18,500 = 26,000 ft3. So S2 + O2*300 = 26,000. If we have stage-outflow table, find O2 such that S2+300 O2 = 26,000. Suppose at stage 5 ft, S=24,000, O=4 cfs: 24,000+1200=25,200. At stage 6 ft, S=30,000, O=8 cfs: 30,000+2400=32,400. Interpolate: 26,000 between 25,200 and 32,400. Fraction = (26,000-25,200)/(32,400-25,200)=800/7200=0.111. O2=4+0.111*(8-4)=4.44 cfs. S2=24,000+0.111*(30,000-24,000)=24,666 ft3. Good.

We can present.

Example 3: Outlet structure: Given pond stage H=4 ft above orifice invert, orifice diameter 12 in, C_d=0.6. Compute Q. Area = π(0.5)^2=0.785 ft2. Q = 0.6*0.785*sqrt(2*32.2*4)=0.471*sqrt(257.6)=0.471*16.05=7.56 cfs. Also weir length 8 ft, C=3.33, head above weir crest = 1 ft, Q=3.33*8*1^{1.5}=26.64 cfs. Total = 34.2 cfs.

Example 4: NRCS runoff depth and volume. 50-acre watershed, CN=75, P=5 in. Q = (5-0.2*S)^2/(5+0.8S). S=1000/75-10=13.333-10=3.333. 0.2S=0.667. Q=(4.333)^2/(5+2.667)=18.78/7.667=2.45 in. Runoff volume = Q/12 * area = 2.45/12 * 50 acres = 10.21 acre-ft. Convert to ft3: 10.21*43560=444,800 ft3.

Example 5: RUSLE. Given R=150, K=0.28, LS=2.5, C=0.06, P=0.5. A=150*0.28*2.5*0.06*0.5 = 3.15 tons/acre-yr.

Need self-test: 15 questions with answers.

Need high-yield checklist.

Need flag unspecified: local design storms, WQv, TR-55 graphical if not supplied.

Let's compose final. Ensure no mention "guaranteed pass". Use "exam-focused preparation designed to maximize readiness." End with disclaimer.

Let's write.

PE Civil: Construction — Detention/Retention Ponds

Exam-Focused Review for the NCEES PE Civil: Construction Exam

Based on NCEES PE Civil Construction exam specifications and the *PE Civil Reference Handbook*, Version 2.0.

Scope: Only Detention/Retention Ponds within Hydraulics and Hydrology. Other topics are included only as needed to explain detention and retention ponds.

> **Disclaimer:** This review is exam-preparation material designed to maximize readiness. It does not guarantee a passing score. Always follow the current NCEES exam specifications and the supplied *PE Civil Reference Handbook* during the exam. If a constant, chart, or local criterion is not supplied by NCEES or in the exam question, do not invent it—flag it and use only what is provided.

0. How to Use This Review

Priority	Topic	Why It Matters
1	Storage volume + critical duration	Core detention design calculation
2	Modified Puls routing	Most likely routing method
3	Outlet hydraulics: orifice, weir, riser	Determines outflow and stage
4	Stage-storage-discharge relationships	Required for routing and sizing
5	Design storms, IDF, t_c, runoff methods	Inputs to pond design
6	Water quality volume + forebay	Common construction/ environmental question
7	Construction, maintenance, safety	Field and design traps

Exam strategy:

- Write units on every line.
- Check whether the question asks for storage volume, peak outflow, or routing.
- For routing, use the correct time step and consistent units.
- For outlets, identify whether flow is orifice-controlled, weir-controlled, or both.

- For storage volume, check multiple storm durations to find the critical one.
- For water quality, use local criteria if supplied; otherwise flag.

1. Purpose and Types

1.1 Definitions and Differences

Feature	Detention Pond	Retention Pond
Storage	Temporarily stores runoff	Stores runoff permanently
Release	Slowly releases through outlet	No continuous release; loses water by infiltration, evaporation, or controlled discharge
Permanent pool	Usually dry between storms	Permanent pool of water
Primary purpose	Peak flow attenuation, quantity control	Water quality, quantity control, recharge, habitat
Also called	Dry detention basin, dry pond	Wet pond, wet detention basin, retention basin
Outlet	Required	May have emergency spillway, but normal pool maintained

Extended detention: Detention basin that holds runoff for an extended period (e.g., 24–48 hours) to improve water quality.

Stormwater wetland: Shallow retention pond with vegetation for treatment.

Infiltration system: Stores runoff and allows it to infiltrate into the ground.

Swale: Vegetated channel for conveyance and treatment; not a pond but related practice.

1.2 Water Quality, Quantity, and Peak Flow Attenuation

- **Quantity control:** Reduce peak discharge to pre-development levels.
- **Water quality control:** Capture and treat the first flush (water quality volume).
- **Peak flow attenuation:** Store excess inflow and release at allowable rate.
- **Recharge:** Infiltration systems and retention ponds can recharge groundwater.

1.3 Common Errors, Assumptions, and Exam Traps

Trap	Correction
Confusing detention and retention	Detention = temporary; retention = permanent pool
Assuming detention pond always dry	Extended detention may have small permanent pool
Forgetting water quality volume	Required by many permits
Using wrong pond type for infiltration	Infiltration systems are not ponds with surface release

> **Flag:** Local criteria define water quality volume, permanent pool depth, and release rates. Use only what is supplied.

2. Hydrology Inputs for Pond Design

2.1 Design Storm Selection and Return Periods

- **Design storm:** Rainfall event used for design (e.g., 2-yr, 10-yr, 25-yr, 100-yr).
- **Return period T:** Average recurrence interval; $T = 1/p$, where p = annual exceedance probability.
- **Risk over n years:** $1 - (1 - 1/T)^n$.
- **Reliability:** $(1 - 1/T)^n$.

Typical design storms:

- Water quality: 1-yr, 24-hr or first flush.
- Channel protection: 1-yr or 2-yr.
- Peak control: 10-yr, 25-yr, 100-yr.
- Emergency spillway: 100-yr or larger.

2.2 Rainfall Intensity–Duration–Frequency (IDF)

$$i = \frac{cT^e}{T_d^m}$$

or

$$i = \frac{cT^e}{(T_d + f)^m}$$

where:

- i = intensity (in./hr)
- T_d = duration (min)
- T = return period (years)
- c, e, f, m = coefficients

Procedure:

1. Determine time of concentration t_c.
2. Set duration $T_d = t_c$ for peak flow.
3. Read i from IDF curve for design return period.

2.3 Time of Concentration t_c

SCS lag formula:

$$t_L = \frac{0.000526(1000/CN - 9)^{0.7} L^{0.8}}{S^{0.5}}$$

$$t_c = \frac{5}{3} t_L$$

where:

- t_L = lag time (hr)
- L = watershed length (ft)
- S = watershed slope (%)
- CN = curve number

Sheet flow formula:

$$T_t = \frac{K_u (nL)^{0.8}}{(P_2)^{0.5} S^{0.4}}$$

- $K_u = 0.933$ (USCS)
- n = Manning's roughness for overland flow
- L = flow length (ft)
- P_2 = 2-yr, 24-hr rainfall (in.)
- S = slope (ft/ft)

Shallow concentrated flow and channel flow:

$$T_t = \frac{L}{V}$$

2.4 Runoff Methods

Rational Method

$$Q = CiA$$

- Q = peak discharge (cfs)

- C = runoff coefficient
- i = rainfall intensity (in./hr) for duration t_c
- A = drainage area (acres)

Composite C:

$$C_w = \frac{\sum C_i A_i}{\sum A_i}$$

NRCS (SCS) Rainfall-Runoff Method

$$Q = \frac{(P - 0.2S)^2}{P + 0.8S}$$

$$S = \frac{1000}{CN} - 10$$

where:

- Q = runoff depth (in.)
- P = precipitation (in.)
- S = maximum basin retention (in.)
- CN = curve number

Initial abstraction: $I_a = 0.2S$.
If $P \leq I_a$, $Q = 0$.

Kinematic Wave Runoff

$$Q = \alpha A^m$$

$$\alpha = \frac{1.486}{n} S_0^{1/2} P_c^{2/3}$$

- $m = 5/3$ for wide rectangular/trapezoidal
- $m = 1.0$ for circular pipe
- $m = 1.33$ for triangular

USGS Regression Method

$$RQ_T = aA^bB^cC^d$$

Region-specific; use only if supplied.

2.5 Unit Hydrograph Application to Pond Inflow

SCS unit hydrograph:

$$q_p = \frac{484AQ}{t_p}$$

$$t_p = \frac{D}{2} + t_L$$

$$T_b = 2.67t_p$$

where:

- q_p = peak flow (cfs)
- A = area (mi²)
- Q = runoff depth (in.)
- t_p = time to peak (hr)
- D = duration of excess rainfall (hr)
- t_L = lag time (hr)

Snyder synthetic unit hydrograph:

$$t_p = C_t(LL_c)^{0.3}$$

$$q_p = \frac{640C_pA}{t_p}$$

Clark unit hydrograph: Time–area method + linear reservoir routing.

2.6 Critical Storm Duration

For detention design, the critical storm duration is the duration that produces the maximum required storage volume. Check multiple durations (e.g., 10, 20, 30, 45, 60 min) and use the largest storage requirement.

Modified Rational Method:

$$V = (Q_{in} - Q_{out}) \times t_d$$

where:

- V = required storage volume (ft³)
- Q_{in} = peak inflow for duration t_d (cfs)
- Q_{out} = allowable outflow (cfs)
- t_d = storm duration (sec)

2.7 Common Errors, Assumptions, and Exam Traps

Trap	Correction
Using t_c for wrong return period	t_c is independent of return period
Forgetting to convert t_d to seconds	Q in cfs × time in sec = ft³
Using wrong CN	Check land use and soil group
Forgetting I_a	$Q = 0$ if $P \leq 0.2S$
Misreading IDF	Match duration to critical storm duration
Using Rational Method for large watershed	Limited to small watersheds, typically < 200 acres

3. Storage Volume Estimation

3.1 Required Storage Volume for Peak Flow Attenuation

Basic equation:

$$V_s = \max(V_{in} - V_{out})$$

where:

- V_s = required storage volume
- V_{in} = inflow volume
- V_{out} = outflow volume

Rational Method storage volume:

$$V = (Q_{in} - Q_{out}) \times t_d$$

- $Q_{in} = CiA$ for duration t_d
- Q_{out} = allowable release rate
- t_d = critical storm duration (sec)

Modified Rational Method:

- Iterate different durations to find maximum V.
- Critical duration occurs when $dV/dt = 0$.

3.2 TR-55 Storage Volume Estimation

TR-55 provides a graphical method:

$$\frac{V_s}{V_r} = f\left(\frac{q_o}{q_i}\right)$$

where:

- V_s = required storage volume
- V_r = runoff volume

- q_o = peak outflow
- q_i = peak inflow

> **Flag:** TR-55 graphs are not fully reproduced in the PE Civil Reference Handbook. If the exam question does not supply the graph or equation, use the Modified Rational Method or the routing method supplied.

3.3 Stage-Storage Relationship

Elevation-area-volume curves:

$$V = \int A(h)\, dh$$

Trapezoidal approximation:

$$V = \frac{A_1 + A_2}{2} \Delta h$$

Prismoidal formula:

$$V = \frac{\Delta h}{6}(A_1 + 4A_m + A_2)$$

where A_m = area at mid-depth.

Common pond shapes:

- Rectangular: $A = LW$
- Trapezoidal: $A = (L + 2zh)(W + 2zh)$
- Circular: $A = \pi r^2$

3.4 Storage Components

Component	Description
Dead storage	Permanent pool; not used for attenuation
Active storage	Used for runoff attenuation; between dead storage and maximum water surface
Freeboard	Vertical distance between maximum water surface and top of embankment
Emergency storage	Storage above primary outlet, below emergency spillway crest

Freeboard: Typically 1–2 ft, or as required by local criteria.

3.5 Common Errors, Assumptions, and Exam Traps

Trap	Correction
Using total pond volume as active storage	Subtract dead storage
Forgetting freeboard	Required for safety
Using wrong t_d	Critical duration must be found by iteration
Assuming linear stage-storage	Use actual elevation-area-volume curve
Forgetting units	V in ft³, acre-ft, or m³

4. Routing Methods

4.1 Level Pool Routing (Modified Puls Method)

Storage equation:

$$\frac{I_1 + I_2}{2}\Delta t - \frac{O_1 + O_2}{2}\Delta t = S_2 - S_1$$

Rearranged for routing:

$$\frac{I_1 + I_2}{2}\Delta t + S_1 - \frac{O_1 \Delta t}{2} = S_2 + \frac{O_2 \Delta t}{2}$$

Standard form:

$$\frac{I_1 + I_2}{2}\Delta t + \left(S_1 - \frac{O_1 \Delta t}{2}\right) = S_2 + \frac{O_2 \Delta t}{2}$$

Procedure:

1. Develop inflow hydrograph $I(t)$.
2. Develop stage-storage curve $S(h)$.
3. Develop stage-discharge curve $O(h)$.
4. Compute $S + O\Delta t/2$ for each stage.
5. At each time step, compute LHS.
6. Find stage where $S + O\Delta t/2 =$ LHS.
7. Read O_2 and S_2.
8. Repeat for next time step.

Tabular routing example:

Time (min)	I (cfs)	$(I_1 + I_2)/2\Delta t$ (ft³)	$S_1 - O_1\Delta t/2$ (ft³)	$S_2 + O_2\Delta t/2$ (ft³)	O_2 (cfs)	S_2 (ft³)
0	10	–	20,000	–	5	20,000
10	15	7,500	18,500	26,000	4.44	24,666
20	25	12,000	24,666 - 4.44*300 = 23,334	35,334	…	…

4.2 Outflow Relationships

Orifice flow:

$$Q = C_d A \sqrt{2gh}$$

- C_d ≈ 0.6 for sharp-edged orifice
- A = orifice area
- h = head above orifice centerline

Weir flow:

$$Q = C L H^{3/2}$$

- Sharp-crested rectangular: $C \approx 3.33$ (USCS)
- Broad-crested: $C \approx 3.0$ (USCS)
- Ogee spillway: $Q = C L H_e^{3/2}$

Riser/standpipe:

- Orifice control at low head
- Weir control at higher head
- Combination: sum of orifice and weir flow

Culvert outlet control:

- Use inlet control and outlet control equations
- Headwater = max(inlet control, outlet control)

4.3 Stage-Discharge Curves

Procedure:

1. Assume a stage h.
2. Compute Q for each outlet structure.

3. Sum all outflows.
4. Plot Q vs. h.

Example:

- Orifice at bottom: $Q_o = C_d A_o \sqrt{2gh}$
- Weir at higher elevation: $Q_w = CL(h - h_w)^{3/2}$ for $h > h_w$
- Total $Q = Q_o + Q_w$

4.4 Common Errors, Assumptions, and Exam Traps

Trap	Correction
Using wrong time step	Δt must be consistent with hydrograph
Forgetting to convert Δt to seconds	Q in cfs, Δt in sec
Neglecting tailwater	Tailwater can submerge outlet
Using wrong weir/orifice coefficient	Match structure type
Forgetting emergency spillway	Required for safety
Using linear interpolation incorrectly	Check stage-storage-discharge table

5. Outlet Structures and Hydraulics

5.1 Primary Outlet Structures

- **Riser:** Vertical pipe with orifice and/or weir openings.
- **Weir:** Sharp-crested, broad-crested, ogee.
- **Orifice:** Circular or rectangular opening.
- **Outlet pipe:** Conveys flow downstream.

- **Emergency spillway:** Auxiliary outlet for extreme storms.

5.2 Weir Flow

Sharp-crested rectangular weir:

$$Q = CLH^{3/2}$$

- $C \approx 3.33$ (USCS)
- L = weir length
- H = head above crest

Sharp-crested with end contractions:

$$Q = C[L - n(0.1H)]H^{3/2}$$

- n = number of end contractions

Broad-crested weir:

$$Q = CLH^{3/2}$$

- $C \approx 3.0$ (USCS)

V-notch weir:

$$Q = CH^{5/2}$$

- $C \approx 2.5$ for 90° V-notch

5.3 Orifice Flow

Free discharge:

$$Q = C_d A\sqrt{2gh}$$

- $C_d \approx 0.6$ for sharp-edged
- h = head above orifice centerline

Submerged discharge:

$$Q = C_d A \sqrt{2g(h_1 - h_2)}$$

- h_1 = upstream head
- h_2 = downstream head

5.4 Culvert Outlet Control

Inlet control:

$$\frac{HW}{D} = K \left(\frac{Q}{AD^{0.5}} \right)^M + K_s S$$

Outlet control:

$$H = H_e + H_f + H_o + H_b + H_j + H_g$$

$$HW = H + TW - LS_0$$

5.5 Energy Dissipation and Outlet Protection

- **Riprap:** Protect against erosion.
- **Stilling basin:** Dissipate energy.
- **Apron:** Prevent scour.
- **Energy dissipator:** Required at outlet.

5.6 Common Errors, Assumptions, and Exam Traps

Trap	Correction
Using wrong C_d	Match orifice type
Forgetting tailwater	Submerged orifice reduces flow
Using wrong weir coefficient	Sharp vs. broad vs. ogee
Forgetting orifice/weir combination	Sum flows
Neglecting energy dissipation	Required for outlet protection
Using wrong head	Head above orifice/weir crest

6. Water Quality and Environmental Considerations

6.1 Water Quality Volume

Water quality volume (WQv):

$$WQ_v = \frac{PAC}{12}$$

where:

- P = first flush depth (in.), typically 0.5 in.
- A = impervious area (acres)
- C = runoff coefficient
- 12 = conversion to acre-ft

> **Flag:** Local criteria define WQv. Use only if supplied.

6.2 Sediment Forebays and Pretreatment

- **Forebay:** Small basin at inlet to capture sediment.
- **Pretreatment:** Vegetated swale, sediment trap, or filter strip.

6.3 Nutrient Removal and Pollutant Reduction

- **Retention ponds:** Settling, biological uptake.
- **Constructed wetlands:** Nutrient uptake by plants.
- **Infiltration:** Pollutants removed by soil.

6.4 Infiltration and Groundwater Recharge

- **Infiltration basin:** Stores runoff, allows infiltration.
- **Permeability:** Soil must have adequate infiltration rate.
- **Clogging:** Sediment can reduce infiltration.

6.5 Erosion and Sediment Control During Construction

- **RUSLE:** $A = RKLSCP$
- **Silt fence:** For sheet flow only.
- **Sediment basin:** Capture sediment.
- **Mulch:** Protect slopes.
- **SWPPP:** Stormwater Pollution Prevention Plan.

6.6 Mosquito and Vector Control

- **Dry detention:** Less mosquito habitat.
- **Wet retention:** May require aeration or mosquito fish.
- **Maintenance:** Remove standing water, control vegetation.

6.7 Common Errors, Assumptions, and Exam Traps

Trap	Correction
Forgetting WQv	Required by many permits
Using wrong first flush depth	Local criteria

Trap	Correction
Ignoring forebay	Required for pretreatment
Using silt fence for concentrated flow	Silt fence is for sheet flow
Forgetting mosquito control	Wet ponds may need management

7. Construction, Maintenance, and Safety

7.1 Site Layout, Excavation, and Embankment Construction

- **Embankment:** Compacted fill, stable side slopes.
- **Side slopes:** Typically 3H:1V to 4H:1V.
- **Liners:** Clay or geomembrane to reduce seepage.
- **Seepage control:** Cutoff trench, toe drain.

7.2 Liners, Soil Permeability, and Seepage Control

- **Permeability:** $k \leq 1 \times 10^{-6}$ cm/s for clay liner.
- **Geomembrane:** HDPE, PVC.
- **Filter criteria:** Prevent piping.

7.3 Inlet and Outlet Protection

- **Riprap:** D_{50} sized for velocity.
- **Apron:** Concrete or riprap.
- **Energy dissipator:** Stilling basin.

7.4 Inspection, Maintenance, and Sediment Removal

- **Inspection:** After each storm.

- **Sediment removal:** When capacity reduced by 10–20%.
- **Vegetation:** Mow, remove invasive species.
- **Outlet maintenance:** Clear debris.

7.5 Safety Requirements

- **Fencing:** Required for safety if deep water.
- **Access:** For maintenance.
- **Embankment stability:** Check slope stability.
- **Warning signs:** For deep water.

7.6 Common Field and Design Errors

- Inadequate storage volume
- Undersized outlet
- Clogged orifice
- Sediment accumulation
- Erosion at outlet
- Poor vegetation establishment
- Inadequate freeboard

8. Worked Examples

Example 1 — Required Storage Volume (Modified Rational Method)

Problem: 15-acre watershed, $C = 0.35$. 25-yr IDF: $i = 4.0$ in./hr for 20 min, $i = 3.2$ in./hr for 30 min, $i = 2.5$ in./hr for 45 min. Allowable outflow = 6 cfs. Find critical duration and required storage.

Solution:

$$Q_{in} = CiA$$

- 20 min: $Q = 0.35(4.0)(15) = 21$ cfs
  $V = (21 - 6)(20 \times 60) = 15 \times 1200 = 18,000$ ft³
- 30 min: $Q = 0.35(3.2)(15) = 16.8$ cfs
  $V = (16.8 - 6)(30 \times 60) = 10.8 \times 1800 = 19,440$ ft³
- 45 min: $Q = 0.35(2.5)(15) = 13.125$ cfs
  $V = (13.125 - 6)(45 \times 60) = 7.125 \times 2700 = 19,237.5$ ft³

Critical duration: 30 min.
Required storage: $V \approx 19,440$ ft³.

Example 2 — Modified Puls Routing

Problem: $\Delta t = 10$ min = 600 sec. At time 1: $S_1 = 20,000$ ft³, $O_1 = 5$ cfs, $I_1 = 10$ cfs. At time 2: $I_2 = 15$ cfs. Stage-storage-discharge: at $h = 5$ ft, $S = 24,000$ ft³, $O = 4$ cfs; at $h = 6$ ft, $S = 30,000$ ft³, $O = 8$ cfs. Find O_2 and S_2.

Solution:

$$\frac{I_1 + I_2}{2}\Delta t + S_1 - \frac{O_1 \Delta t}{2} = S_2 + \frac{O_2 \Delta t}{2}$$

$$\frac{10 + 15}{2}(600) + 20,000 - \frac{5(600)}{2} = S_2 + \frac{O_2(600)}{2}$$

$$12.5(600) + 20,000 - 1,500 = 7,500 + 18,500 = 26,000$$

So $S_2 + 300 O_2 = 26,000$.

Interpolate:

- At $h = 5$: $24,000 + 300(4) = 25,200$
- At $h = 6$: $30,000 + 300(8) = 32,400$

Fraction = $(26,000 - 25,200)/(32,400 - 25,200) = 800/7,200 = 0.111$.

$$O_2 = 4 + 0.111(8 - 4) = 4.44 \text{ cfs}$$

$$S_2 = 24,000 + 0.111(30,000 - 24,000) = 24,666 \text{ ft}^3$$

Answer: $O_2 = 4.44$ cfs, $S_2 = 24,666$ ft³.

Example 3 — Orifice and Weir Outflow

Problem: Pond stage $H = 4$ ft above orifice invert. Orifice diameter = 12 in, $C_d = 0.6$. Weir length = 8 ft, crest at 3 ft above orifice invert, $C = 3.33$. Find total outflow.

Solution:
Orifice area:

$$A_o = \pi(0.5)^2 = 0.785 \text{ ft}^2$$

$$Q_o = 0.6(0.785)\sqrt{2(32.2)(4)} = 0.471\sqrt{257.6} = 0.471(16.05) = 7.56 \text{ cfs}$$

Weir head: $H_w = 4 - 3 = 1$ ft.

$$Q_w = 3.33(8)(1)^{3/2} = 26.64 \text{ cfs}$$

Total:

$$Q = 7.56 + 26.64 = 34.2 \text{ cfs}$$

Example 4 — NRCS Runoff Volume

Problem: 50-acre watershed, $CN = 75$, $P = 5$ in. Find runoff depth and volume.

Solution:

$$S = \frac{1000}{75} - 10 = 13.333 - 10 = 3.333 \text{ in.}$$

$$Q = \frac{(5 - 0.2(3.333))^2}{5 + 0.8(3.333)} = \frac{(5 - 0.667)^2}{5 + 2.667} = \frac{4.333^2}{7.667} = \frac{18.78}{7.667} = 2.45 \text{ in.}$$

Runoff volume:

$$V = \frac{Q}{12}A = \frac{2.45}{12}(50) = 10.21 \text{ acre-ft}$$

$$V = 10.21(43,560) = 444,800 \text{ ft}^3$$

Example 5 — RUSLE Erosion

Problem: $R = 150$, $K = 0.28$, $LS = 2.5$, $C = 0.06$, $P = 0.5$. Find annual soil loss.

Solution:

$$A = RKLSCP = 150(0.28)(2.5)(0.06)(0.5) = 3.15 \text{ tons/acre-yr}$$

9. High-Yield Checklist

□ Detention vs. retention definitions

□ $T = 1/p$, risk = $1 - (1 - 1/T)^n$

□ IDF: $i = cT^e / T_d^m$

□ Rational Method: $Q = CiA$

□ NRCS: $Q = (P - 0.2S)^2/(P + 0.8S)$, $S = 1000/CN - 10$

□ t_c: SCS lag, sheet flow, channel flow

□ Storage volume: $V = (Q_{in} - Q_{out}) \times t_d$

□ Modified Puls routing: $(I_1 + I_2)/2\Delta t + S_1 - O_1\Delta t/2 = S_2 + O_2\Delta t/2$

□ Stage-storage: $V = \int A\, dh$

□ Stage-discharge: $Q_o = C_d A\sqrt{2gh}$, $Q_w = CLH^{3/2}$

□ Riser/standpipe: orifice + weir combination

□ Emergency spillway and freeboard

□ Water quality volume

□ Sediment forebay

□ RUSLE: $A = RKLSCP$

□ Silt fence for sheet flow only

□ SWPPP

□ Maintenance and safety

□ Units, units, units

10. Self-Test

1. What is the difference between detention and retention?
2. What is the return period for a storm with 2% annual exceedance probability?
3. What is the risk of a 50-year storm over 20 years?
4. What is the Rational Method formula?
5. $P = 6$ in., $CN = 80$. Find Q.
6. What is the SCS lag formula?
7. What is the storage volume equation using Modified Rational Method?
8. What is the Modified Puls routing equation?
9. What is the orifice flow equation?
10. What is the sharp-crested weir equation?
11. What is the broad-crested weir equation?
12. What is the V-notch weir equation?
13. What is the RUSLE equation?
14. What is the water quality volume?
15. What is the critical storm duration?
16. What is the difference between dead storage and active storage?
17. What is freeboard?
18. What is the purpose of a sediment forebay?
19. What is the purpose of an emergency spillway?
20. What is the difference between inlet control and outlet control for culverts?

Answers:

1. Detention temporarily stores; retention stores permanently.
2. $T = 1/0.02 = 50$ years.
3. Risk = $1 - (1 - 1/50)^{20} = 1 - 0.98^{20} = 1 - 0.6676 = 0.332 = 33.2\%$.
4. $Q = CiA$.
5. $S = 1000/80 - 10 = 2.5$; $Q = (6 - 0.5)^2/(6 + 2) = 30.25/8 = 3.78$ in.
6. $t_L = 0.000526(1000/CN - 9)^{0.7} L^{0.8}/S^{0.5}$.
7. $V = (Q_{in} - Q_{out})t_d$.
8. $(I_1 + I_2)/2\Delta t + S_1 - O_1\Delta t/2 = S_2 + O_2\Delta t/2$.
9. $Q = C_d A\sqrt{2gh}$.
10. $Q = CLH^{3/2}$.
11. $Q = CLH^{3/2}$ (with $C \approx 3.0$).
12. $Q = CH^{5/2}$.
13. $A = RKLSCP$.
14. $WQ_v = PAC/12$.
15. Duration that produces maximum required storage.
16. Dead storage = permanent pool; active storage = used for attenuation.
17. Vertical distance between maximum water surface and top of embankment.
18. Capture sediment and pretreat inflow.
19. Auxiliary outlet for extreme storms to protect embankment.
20. Inlet control: inlet geometry controls; outlet control: outlet/tailwater/friction controls.

Final Note: This review is aligned with the NCEES PE Civil: Construction exam specifications and the *PE Civil Reference Handbook*, Version 2.0. If a detail is not specified by NCEES or falls outside the stated scope, it has been flagged. Use this material as exam-focused preparation designed to maximize readiness; it does not guarantee a passing score.

> generate mock exam after actual exams questions covering the Hydraulics and Hydrology

Thought for 6 seconds

The user wants a mock exam based on actual NCEES PE Civil: Construction exam questions covering the Hydraulics and Hydrology area. I need to generate realistic practice problems. To do this, I should search for actual exam questions, sample problems, and practice exams related to the Hydraulics and Hydrology topics on the PE Civil Construction exam. I'll search for terms like "NCEES PE Civil Construction Hydraulics and Hydrology practice questions," "PE Civil Construction stormwater practice problems," and "PE Civil Construction hydrology sample questions." I'll also look for specific topics like open-channel flow, culverts, storm sewers, and detention ponds.

Found 44 web pages

The search results include some potentially useful links. I'll open some of them to gather sample questions and problems.

Read 5 pages PE Genius | Civil PE Exam Preperation ↗

PE Civil Construction Drainage & Stormwater Exam 2026–2027 | 100 Challenging Practice Ques | Exams Civil Eng... ↗

Practice Problems Hydrology - Civil Engineering (PE Civil) - PE Exam - PE Exam Exam > PE Exam Notes > Civil En... ↗

Practice Problems Hydraulics - Civil Engineering (PE Civil) - PE Exam - PE Exam Exam > PE Exam Notes > Civil E... ↗

PE Genius | Civil PE Exam Preperation ↗

The search results provide some sample questions. I'll use these to create a mock exam. I'll also search for more specific topics like "PE Civil Construction hydrology practice questions PDF" and "PE Civil Construction hydraulics practice problems" to gather more material.

Found 19 web pages

The search results provide a variety of sample questions and problems. I'll now create a mock exam with questions covering the Hydraulics and Hydrology topics. I'll organize the exam into sections: Open-Channel Flow, Culverts, Storm Sewers, Watersheds, Detention/Retention Ponds, and Erosion Control. I'll include multiple-choice questions and some calculation problems, with detailed solutions. I'll also include some questions based on the sample problems found. I'll ensure the questions are realistic and aligned with the exam specifications. I'll also include a high-yield checklist and self-test at the end.

PE Civil: Construction — Mock Exam

Hydraulics and Hydrology: Stormwater Collection, Drainage, and Detention/Retention Ponds

Instructions:

- This mock exam contains 25 multiple-choice questions and 5 calculation problems.
- Time limit: 2 hours.

- Use the PE Civil Reference Handbook, Version 2.0, as needed.
- All questions are exam-focused and aligned with NCEES PE Civil: Construction specifications.
- Show all work for calculation problems.
- Unit consistency is critical—check USCS vs. SI.

> **Disclaimer:** This mock exam is exam-preparation material designed to maximize readiness. It does not guarantee a passing score. If a detail is not specified by NCEES or falls outside the stated scope, it has been flagged.

Section 1: Multiple-Choice Questions (25 Questions, 1 Point Each)

Open-Channel Flow

1. A trapezoidal channel has bottom width $b = 5$ ft, side slopes $z = 2$ H:1 V, flow depth $y = 4$ ft. What is the cross-sectional area of flow?
A. 20 ft²
B. 36 ft²
C. 52 ft²
D. 64 ft²

2. For a rectangular channel with discharge per unit width $q = 12$ cfs/ft, what is the critical depth?
A. 1.24 ft
B. 1.65 ft
C. 2.05 ft
D. 2.47 ft

3. A hydraulic jump occurs in a rectangular channel with upstream depth $y_1 = 1.2$ ft and upstream Froude number $Fr_1 = 3.0$. What is the downstream depth y_2?
A. 2.85 ft
B. 3.42 ft
C. 4.18 ft
D. 5.06 ft

4. Which of the following GVF profiles represents a backwater curve on a mild slope?

A. M1

B. M2

C. M3

D. S1

5. In a circular pipe flowing partially full, maximum discharge occurs at approximately what d/D ratio?

A. 0.50

B. 0.81

C. 0.94

D. 1.00

Culverts

6. A culvert is classified as USGS Type 4 when:

A. Inlet is submerged, outlet is not submerged

B. Both inlet and outlet are submerged

C. Inlet is not submerged, outlet is submerged

D. Neither inlet nor outlet is submerged

7. For a circular concrete culvert with square edge and headwall, the inlet control unsubmerged constants are:

A. $K = 0.0098$, $M = 2.0$

B. $K = 0.0018$, $M = 2.0$

C. $K = 0.0078$, $M = 2.0$

D. $K = 0.0210$, $M = 1.33$

8. A culvert has $HW_i = 8.5$ ft and $HW_o = 6.2$ ft. What is the controlling headwater depth?

A. 6.2 ft, outlet control

B. 8.5 ft, inlet control

C. 14.7 ft, combined control

D. 2.3 ft, differential control

9. The entrance loss coefficient k_e for a concrete pipe projecting from fill with a socket end is:

A. 0.2

B. 0.5

C. 0.7

D. 0.9

10. In outlet control, the total head loss equation is:

A. $H = H_e + H_f + H_o$

B. $H = H_e + H_f + H_o + H_b + H_j + H_g$

C. $H = H_f + H_o$

D. $H = H_e + H_o$

Storm Sewers

11. The Rational Method formula $Q = CiA$ gives Q in:

A. cfs when i is in in./hr and A is in acres

B. cfs when i is in in./hr and A is in ft²

C. gpm when i is in in./hr and A is in acres

D. cfs when i is in cm/hr and A is in hectares

12. A 25-acre watershed has a composite runoff coefficient of 0.65. The rainfall intensity for the design storm is 4.2 in./hr. What is the peak runoff rate?

A. 17.75 cfs

B. 68.25 cfs

C. 105.0 cfs

D. 42.5 cfs

13. For a triangular gutter with cross slope $S_x = 0.02$, longitudinal slope $S_L = 0.01$, Manning's $n = 0.016$, and spread $T = 8$ ft, what is the flow rate?

A. 0.85 cfs

B. 1.31 cfs

C. 2.05 cfs

D. 3.42 cfs

14. The minimum velocity in a storm sewer is typically:

A. 1 ft/s

B. 2 ft/s

C. 5 ft/s

D. 10 ft/s

15. The hydraulic grade line (HGL) in a storm sewer must remain:

A. Above the pipe crown

B. Below the gutter line

C. At the pipe invert

D. Above the energy grade line

Watersheds and Hydrology

16. A 10-year storm has an annual exceedance probability of:

A. 1%

B. 5%

C. 10%

D. 20%

17. The risk of a 25-year storm occurring at least once in 10 years is most nearly:

A. 10%

B. 25%

C. 33%

D. 40%

18. For $CN = 80$ and $P = 4$ in., the NRCS runoff depth Q is most nearly:

A. 1.2 in.

B. 2.0 in.

C. 2.8 in.

D. 3.5 in.

19. The SCS lag formula gives t_L in:

A. Minutes

B. Hours

C. Seconds

D. Days

20. The Snyder synthetic unit hydrograph peak flow equation is:

A. $q_p = 484AQ/t_p$

B. $q_p = 640C_pA/t_p$

C. $q_p = 1.49AR^{2/3}S^{1/2}/n$

D. $q_p = CiA$

Detention/Retention Ponds

21. The difference between a detention pond and a retention pond is:

A. Detention ponds have a permanent pool; retention ponds do not

B. Retention ponds have a permanent pool; detention ponds do not

C. Detention ponds are only for water quality; retention ponds are only for quantity

D. There is no difference

22. The Modified Puls routing equation is:

A. $V = (Q_{in} - Q_{out})t$

B. $\frac{I_1+I_2}{2}\Delta t + S_1 - \frac{O_1\Delta t}{2} = S_2 + \frac{O_2\Delta t}{2}$

C. $Q = CiA$

D. $S = \frac{1000}{CN} - 10$

23. The orifice flow equation is:

A. $Q = CLH^{3/2}$

B. $Q = C_dA\sqrt{2gh}$

C. $Q = CH^{5/2}$

D. $Q = \frac{1.486}{n}AR^{2/3}S^{1/2}$

24. The water quality volume is typically calculated using:

A. The 100-year storm

B. The first flush depth (e.g., 0.5 in.)

C. The peak discharge

D. The time of concentration

25. Freeboard in a detention pond is:

A. The permanent pool depth

B. The vertical distance between maximum water surface and top of embankment

C. The active storage volume

D. The emergency spillway capacity

Section 2: Calculation Problems (5 Problems, 5 Points Each)

Problem 1 — Open-Channel Flow (Manning's Equation)

Problem: A trapezoidal channel has bottom width $b = 6$ ft, side slopes $z = 2$ H:1 V, Manning's $n = 0.025$, and longitudinal slope $S = 0.001$. The flow rate is $Q = 120$ cfs. Find the normal depth y_n.

Solution:

$$A = (b + zy)y = (6 + 2y)y$$

$$P = b + 2y\sqrt{1 + z^2} = 6 + 2y\sqrt{5} = 6 + 4.472y$$

$$R_H = A/P$$

$$Q = \frac{1.486}{n} A R_H^{2/3} S^{1/2}$$

Trial $y = 3.25$ ft:

$$A = (6 + 6.5)(3.25) = 40.625 \text{ ft}^2$$

$$P = 6 + 4.472(3.25) = 20.53 \text{ ft}$$

$$R_H = 40.625/20.53 = 1.979 \text{ ft}$$

$$R_H^{2/3} = 1.576$$

$$Q = \frac{1.486}{0.025}(40.625)(1.576)(0.001)^{0.5} = 59.44 \times 40.625 \times 1.576 \times 0.03162 \approx 119.9 \text{ cfs}$$

Answer: $y_n \approx 3.25$ ft.

Problem 2 — Culvert Inlet Control

Problem: A circular concrete culvert has $D = 3$ ft, $Q = 120$ cfs, and a square edge with headwall. Use HDS-5 constants: $K = 0.0098$, $M = 2.0$, $c = 0.0398$, $Y = 0.67$. Find the inlet-control headwater depth.

Solution:

$$A = \frac{\pi D^2}{4} = \frac{\pi (3)^2}{4} = 7.069 \text{ ft}^2$$

$$\frac{Q}{AD^{0.5}} = \frac{120}{7.069\sqrt{3}} = \frac{120}{12.24} = 9.80$$

$$\left(\frac{Q}{AD^{0.5}}\right)^2 = 96.0$$

Unsubmerged:

$$\frac{HW}{D} = 0.0098(96.0) = 0.941$$

$$HW = 0.941(3) = 2.82 \text{ ft}$$

Submerged:

$$\frac{HW}{D} = 0.0398(96.0) + 0.67 = 3.82 + 0.67 = 4.49$$

$$HW = 4.49(3) = 13.47 \text{ ft}$$

Since unsubmerged gives $HW/D < 4$, use unsubmerged.

Answer: $HW_i \approx 2.82$ ft.

Problem 3 — Rational Method and Storm Sewer Design

Problem: A 25-acre watershed has a composite runoff coefficient $C = 0.65$. The time of concentration is 20 min. The 10-year IDF gives $i = 4.2$ in./hr for 20 min. Find the peak flow.

Solution:

$$Q = CiA = 0.65(4.2)(25) = 68.25 \text{ cfs}$$

Answer: $Q \approx 68.3$ cfs.

Problem 4 — Detention Pond Storage Volume (Modified Rational Method)

Problem: A 15-acre watershed has $C = 0.35$. The 25-year IDF gives: $i = 4.0$ in./hr for 20 min, $i = 3.2$ in./hr for 30 min, $i = 2.5$ in./hr for 45 min. The allowable outflow is 6 cfs. Find the critical duration and required storage volume.

Solution:

$$Q_{in} = CiA$$

- 20 min: $Q = 0.35(4.0)(15) = 21$ cfs
  $V = (21 - 6)(20 \times 60) = 15 \times 1200 = 18,000$ ft³
- 30 min: $Q = 0.35(3.2)(15) = 16.8$ cfs
  $V = (16.8 - 6)(30 \times 60) = 10.8 \times 1800 = 19,440$ ft³
- 45 min: $Q = 0.35(2.5)(15) = 13.125$ cfs
  $V = (13.125 - 6)(45 \times 60) = 7.125 \times 2700 = 19,237.5$ ft³

Critical duration: 30 min.
Required storage: $V \approx 19,440$ ft³.

Problem 5 — NRCS Runoff and RUSLE Erosion

Problem: A 50-acre watershed has $CN = 75$ and receives $P = 5$ in. of rainfall.
(a) Find the runoff depth Q.
(b) If $R = 150$, $K = 0.28$, $LS = 2.5$, $C = 0.06$, $P = 0.5$, find the annual soil loss using RUSLE.

Solution (a):

$$S = \frac{1000}{75} - 10 = 3.333 \text{ in.}$$

$$Q = \frac{(5 - 0.2(3.333))^2}{5 + 0.8(3.333)} = \frac{(5 - 0.667)^2}{5 + 2.667} = \frac{4.333^2}{7.667} = \frac{18.78}{7.667} = 2.45 \text{ in.}$$

Solution (b):

$$A = RKLSCP = 150(0.28)(2.5)(0.06)(0.5) = 3.15 \text{ tons/acre-yr}$$

Answer: (a) $Q = 2.45$ in.; (b) $A = 3.15$ tons/acre-yr.

Section 3: Answer Key

Multiple-Choice Answers

Question	Answer	Question	Answer
1	C	14	B
2	B	15	B
3	C	16	C
4	A	17	C
5	C	18	B
6	B	19	B
7	A	20	B

Question	Answer	Question	Answer
8	B	21	B
9	A	22	B
10	B	23	B
11	A	24	B
12	B	25	B
13	B		

Section 4: High-Yield Checklist

□ Manning's equation: USCS vs. SI constant

□ $R_H = A/P$

□ $Fr = V/\sqrt{gD}$, $D = A/T$

□ Critical depth rectangular: $y_c = (q^2/g)^{1/3}$

□ Hydraulic jump: $y_2/y_1 = 0.5(\sqrt{1 + 8Fr_1^2} - 1)$

□ GVF profiles: M, S, C, H, A

□ Circular pipe partial flow: max Q at $d/D \approx 0.94$

□ USGS culvert Types 1–6

□ Inlet control: unsubmerged vs. submerged

□ Outlet control: $H = H_e + H_f + H_o + H_b + H_j + H_g$

□ $HW = \max(HW_i, HW_o)$

□ Rational Method: $Q = CiA$

□ Composite C

□ NRCS: $Q = (P - 0.2S)^2/(P + 0.8S)$, $S = 1000/CN - 10$

□ t_c: sheet flow, SCS lag, Kirpich

□ IDF: i for t_c and return period

□ Curb/gutter: $Q = (0.56/n) S_x^{1.67} S_L^{0.5} T^{2.67}$

□ Detention routing: Modified Puls

□ RUSLE: $A = RKLSCP$

□ HGL below gutter line

□ Minimum/maximum velocities

□ Units, units, units

Section 5: Self-Test

1. What is the USCS Manning constant?
2. For a rectangular channel, $q = 15$ cfs/ft. Find y_c.
3. $Fr = 2.0$, $y_1 = 1.5$ ft. Find y_2.
4. What is the maximum discharge ratio in a circular pipe?
5. What are the USGS culvert Types 1–6?
6. When does inlet control govern?
7. What is the Rational Method formula?
8. $P = 5$ in., $CN = 75$. Find Q.
9. What is the SCS lag formula?
10. What is the gutter flow equation for a triangular gutter?
11. What is the Modified Puls routing equation?
12. What is RUSLE?
13. What is the minimum velocity in a storm sewer?
14. What is the difference between detention and retention?
15. What is the Froude number for a rectangular channel?
16. What is the specific energy equation?
17. What is the hydraulic radius?
18. What is the critical depth equation for a rectangular channel?

19. What is the energy loss in a hydraulic jump?
20. What is the composite Manning's n formula?

Answers:

1. 1.486
2. $y_c = (15^2/32.2)^{1/3} = 1.91$ ft
3. $y_2 = 0.5(\sqrt{1+8(4)}-1)(1.5) = 0.5(5.74)(1.5) = 4.31$ ft
4. $Q/Q_{\text{full}} \approx 1.08$ at $d/D \approx 0.94$
5. Type 1: steep, inlet control; Type 2: mild, outlet control; Type 3: tranquil, outlet control; Type 4: submerged outlet, outlet control; Type 5: rapid inlet, inlet control; Type 6: full free outfall, inlet control
6. When $HW_i > HW_o$
7. $Q = CiA$
8. $S = 1000/75 - 10 = 3.33$; $Q = (5-0.666)^2/(5+2.667) = 18.78/7.667 = 2.45$ in.
9. $t_L = 0.000526(1000/CN - 9)^{0.7} L^{0.8}/S^{0.5}$
10. $Q = (0.56/n) S_x^{1.67} S_L^{0.5} T^{2.67}$
11. $(I_1 + I_2)\Delta t/2 + (S_1/\Delta t - O_1/2) = S_2/\Delta t + O_2/2$
12. Revised Universal Soil Loss Equation: $A = RKLSCP$
13. 2 ft/s
14. Detention temporarily stores and releases; retention stores permanently
15. $Fr = V/\sqrt{gy}$
16. $E = y + V^2/2g$
17. $R_H = A/P$
18. $y_c = (q^2/g)^{1/3}$
19. $\Delta E = (y_2 - y_1)^3/(4y_1y_2)$
20. $n_c = [\sum(p_i n_i^{1.5})/P]^{2/3}$

Final Note: This mock exam is aligned with the NCEES PE Civil: Construction exam specifications and the *PE Civil Reference Handbook*, Version 2.0. If a detail is not specified by NCEES or falls

outside the stated scope, it has been flagged. Use this material as exam-focused preparation designed to maximize readiness; it does not guarantee a passing score.

60 web pages